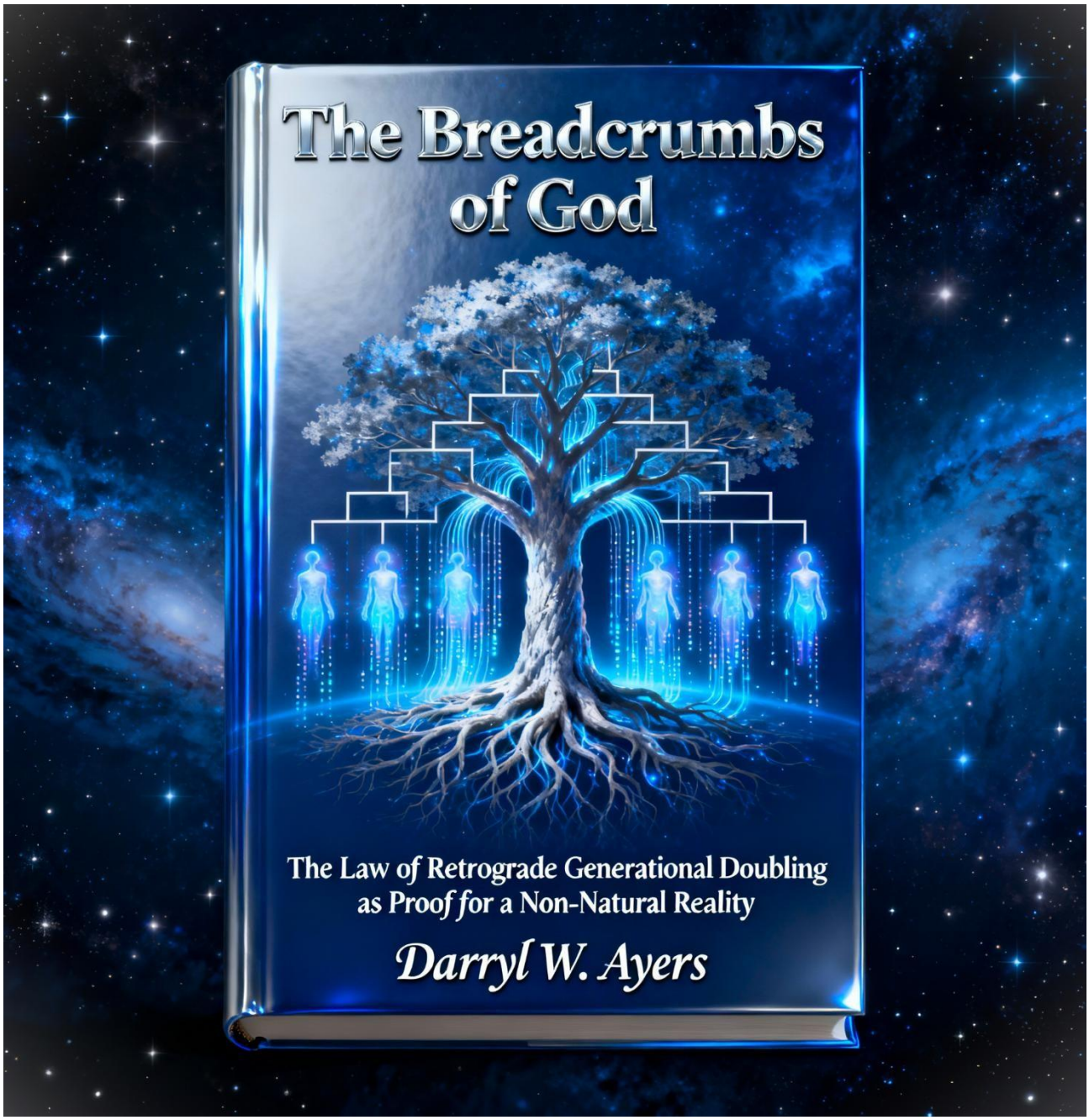
The Breadcrumbs of God
The Law of Retrograde Generational Doubling as Proof for a Non-Natural Reality
Darryl W. Ayers

THE BREADCRUMBS OF GOD

OR:

THE LAW OF RETROGRADE GENERATIONAL DOUBLING
A MATHEMATICAL PROOF FOR NON-NATURAL REALITY.

BY

Darryl W. Ayers

ABSTRACT

This study presents mathematical proof demonstrating fundamental inconsistencies between genealogically mandated ancestor populations and historically estimated demographic data. Through exponential ancestry schematics modeled alongside comprehensive analysis and application of Pedigree Collapse modifiers we establish that conventional understanding of human population dynamics over the past 2,500 years yields mathematically impossible requirements. We propose that these paradoxes necessitate consideration of alternative ontological frameworks for understanding historical continuity, demographic timelines and human propagation dynamics.

***Keywords** population genetics, ancestor paradox, genealogical mathematics, pedigree collapse, demographic paradox, ancestry modeling, retrograde doubling, heredity, ancestral trees, population modeling, demographics, simulation theory, Common Existential Paradigm, artificial construct. Intelligent design, non-natural reality, artificial reality, constructed reality, consciousness development, universe*

TABLE OF CONTENTS

I. LOGIC, MATH & TIME.....5

II. WHAT IS PEDIGREE COLLAPSE.....9

III. THE ROBIN FOX CITATION CONSPIRACY.....12

IV. THE MISAPPROPRIATION OF CHANG STUDIES.....22

V. ONLY METHODOLOGICALLY SOUND RESEARCH:
ROBERT C. GUNDERSON'S EMPIRICAL INTEGRITY.....26

VI. MATHEMATICAL MODELING
TESTING PEDIGREE COLLAPSE SCENARIOS.....33

VII. CASE STUDY: HOW ACADEMIC PSEUDOSCIENCE
PERPETUATES PEDIGREE COLLAPSE MYTHOLOGY.....37

VIII. THE RULE OF INSUFFICIENT CAUSES
THE END OF NATURALISTIC EXPLANATION.....60

IX. THE RECURSIVE ARCHITECTURE OF HUMAN EMERGENCE
WITHIN THE C.E.P.....64

X. FORMAL SUMMATION: THE ANCESTRAL PARADOX, EXISTENTIAL BANDWIDTH AND INSUFFICIENT CAUSES SYLLOGISM

XI. THE GREATEST NEWS IN HUMAN HISTORY.....69

PREFACE

This is not a First Cause, or who came first the chicken or the egg type argument. In fact, this dissertation will concern itself with nothing that has occurred on the alleged planet earth any further back than two thousand five hundred alleged years ago.

INTRODUCTION

You are in this publication. You and every human being to have lived on planet earth for the last 2500 years. You, them and the quadrillions who had to have existed but could not possibly have done so are all represented herein, but that is not the greatest feat of this work.

Phillip Jose Farmer created a fictional literary universe where every human being to have ever existed were all resurrected together at one time along the shores of a 10,000-milelong river, bounded by impassable mountains. His ***River World*** Series chronicles the exploits of a party of intrepid explorers who went in search of the headwaters of this fantastic world to find the parties who were responsible for bringing everyone back to life in the valley formed by this massive water course, Mark Twain and Genghis Khan among them.

Had Mr. Farmer endeavored to speculate on the size of that population, maybe the best kept in plain sight secret of human history would have come to our attention in the 1970's instead of now. This secret is the most revolutionary insight of our age and can now be theoretically and mathematically established beyond all reasonable doubt. The clues left by the Creator(s) leading directly to it have been stumbled across by a simple everyman who just followed the trail of breadcrumbs to the back door of reality. Mankind's next and

possibly final frontier lies before us. The factual existence of a Meta-Universe that has both manufactured and maintains containment of ours may now be rationally postulated, possibly making the threshold to the EXIT from our Common Existential Paradigm discoverable. Who dares to step across with him?

I LOGIC, MATH & TIME

We humans all exist together within the confines of a shared mutual sensory and perceptual universe. It is this Universe I ask leave to henceforth call our Common Existential Paradigm (C.E.P.). The shared or non-shared mutuality of said universe has been thoroughly addressed by others brighter than I, but to counter the elimination of a certain amount of necessary agreement between writer and reader on this point that would require extensive preambles that exist elsewhere, I am submitting the presentation of this theory in the second person. The purpose of this grammatical choice is so that each reader may conceptualize the ideas contained herein for themselves. The truths established herein are universally applicable regardless of station or perspective, as all theorems worth their weight should be.

Adam, Eve, all first-generation homo sapiens and Christ excepted, all other real human beings ever conceived or born on planet Earth came to exist as the result of two other distinctly individual real adult human beings coupling to produce a child. As the preceding is the crux of our argument and the shoulders upon which our thesis is carried, please allow the writer to be both redundant and precise about establishing this self-evident fact.

Every real, extant, non-theoretical, human being existing on planet earth today or ever having previously existed came about by the same exact means. That means being the procreative coupling of two distinctly separate individual human beings of child conceiving and bearing age and ability, and these two always being one male and one female.

The idea of a CHILDLESS human being is easily and entirely conceivable. It is a common happenstance within the framework of our Common Existential Paradigm (C.E.P.). Childless humans are known to all of us either by personal experience or observation of others who have so existed within range of our knowledge and/or common sense. The ease with which any one of us can imagine a childless human being is by converse an even more fully INCONCEIVABLE concept than that anyone could ever possibly be a parentless human. Childless human beings happen every day so to speak, parentless human beings happen not at all.

We are not concerned with the qualitative aspects of said parentage, but with the biologically mandatory quantity thereof. The presence or absence of either parent, the nurturing care or lack thereof from either parent, May December romances, incestuous couplings, prenatal death of the father or postnatal death of the mother, good parents, bad parents or any other conceivable, qualitative aspect of said parentage notwithstanding; all human beings to have ever existed on planet Earth came about as a result of one real human sperm donor and one real human egg donor finding occasion to join the twain and make one.

Two pre-existing parents to account for every human birth in history is as immutable a Law of our existence as there is. If we have not unequivocally garnered the assent of all readers on this point at this point, we do not know further how to obtain it.

Having established the already self-evident fact of the necessity for there to be two extant real living human beings for any one other human being to have ever been produced on earth beyond all contradiction, let us now address another genealogical/demographic phenomenon that affects the plan and purpose of this dissertation.

Generational time cycles are a necessary element of any demographically related study that wishes to examine time frames for populations of antiquity based upon genealogical mandates of human reproduction and propagation in real world environments. Wang, Richard J. et al (2023) in *Science Advances,* established in his study that human generational cycles were recurring at a rate of every 26.9 years going back into antiquity. (1) Fenner, Jack (2005) in the "*American Journal of Anthropology*" suggested that maternal generational intervals occurred at 25 to 28 years. (2) Based on the findings of these two published researchers we have chosen 27.5 years as a base value when seeking to calculate the number of generations over any given historical time frame.

Armed with the inarguable biological fact of two parents for every child, coupled with the utilization of 27.5 years as an average time frame in which generations will cycle, we may proceed. It is with these two very simple self-evident observable and immutable facts of human existence within our Common Existential Paradigm that we may set forth a concise and applicable law, a law that we ask leave to henceforth call the Law of Retrograde Generational Doubling.

The Law of Retrograde Generational Doubling states:

If you postulate the existence of any real, non-theoretical, human being on planet Earth you must also postulate the existence of that human's two equally real nontheoretical progenitors and do so iteratively by generation at average time intervals of 27.5 years.

The law can be expressed as a mathematical formula thusly,

$$ABG=2^{(HTF/AGC)}$$

With
ABG being, Ancestors By Generation
HTF being, Historical Time Frame
AGC being, Average Generational Cycle (in this case 27.5 years)

Let us now examine this law in operation in a normative human environment. We are Gen zero, a single living human being. Our parents are Gen 1, and like every other human being to have ever existed we have two of them, whether we knew them or not. Whether they're still alive or not, they both were alive at one time. As viable reproductive human agents, 1 male 1 female.

All other considerations aside, that's a fact. Now because that's a fact, and since we've now postulated our real two parents we must therefore now postulate their equally real non-theoretical 2 parents each. These are the four grandparents of ours. We are now faced with the necessary demands of the law as well as the demands of common sense. The Law tells us that each of our grandparents must have also had two very real living human beings as their progenitors, and in all normative situations, there are 8 of them, our great grandparents. This continues so on and so forth ad infinitum all the way back into the prehistoric mists of time from where man allegedly arose, but we promised we would not be concerning ourselves with anything that has happened more than 2500 years into the alleged past.

We see here a very simple and straightforward geometrical progression that must repeat every 27 and a 1/2 years on average, along each of your genealogical lines of ascent. This very simple, elegant and undeniable geometric progression continues unabated, and remains continuously in operation, for as long as our ancestors continue to produce parents in any genealogical line of ascent, (something which they must continuously do along all lines otherwise we eradicate our own individual existences). Lines of heredity do terminate, but only when GOING DOWN/FORWARD THROUGH TIME. A line of heredity can only “terminate” when someone on it fails to produce an offspring. For a line of

heredity to terminate in retrograde, a child would have had to have failed to produce parents.

The mandates of the L.R.G.D. (under normal genealogical and demographic conditions) carry with them very profound mathematical consequences. When properly applied the L.R.G.D. mandates the existence of, 2^{50}, or 1.126×10^{15}, or 1,126,000,000,000,000, or one quadrillion, one hundred twenty-six trillion, real, non-theoretical, living together contemporaneously human beings on planet Earth at just 1375 years ago, or 650 C.E. If we extrapolate out to our self-imposed limit of 2500 years ago the number grows to 1.24×10^{24} or 1.24 septillion. This number of coexisting human bodies in one generation would exceed the atomic mass of planet Earth.

These numbers are of course the absurdity of all absurdities. We need not go into very much detail as to why these numbers are absurd, but we must unfortunately touch somewhat extensively upon the matter, because as absurd as they are, they are just as much mandatorily demanded by the immutable laws of biology and mathematics as given to us by our C.E.P., and represent the foundation upon which the writer intends to postulate a non-natural reality.

II WHAT IS PEDIGREE COLLAPSE

The fact of 2 parents is an unceasing reality that can never be challenged, worked around or denied by any rational observer. Two individual and unique parents for each of us is a must, but does this mean that four individual and unique grandparents, 8 individual and unique great-grandparents or 16 individual and unique GG-grandparents carries with it an equivalent measure of “must be”? The answer to this question is in fact, “No”.

Incestuous and closely related couplings that produce offspring; such as those between first cousins, half-siblings and even rarely full siblings, or when uncle's marry nieces or Aunts marry nephews or in cases of polygamy we see a retardation or reduction in the number of direct ***unique individual*** ancestors on the generational tiers usually no less than 3 to 4 generations back from the offspring of such pairings. This genealogical or demographic phenomenon has a name, “Pedigree Collapse”

In 1980, Robert C. Gunderson first propounded the concept of Pedigree Collapse at the World Conference on Records in Salt Lake City, Utah. Robert C. Gunderson (6 December 1931 – 23 June 2003), was an American genealogist. He was the first supervisor of the Royalty Identification Unit of the Genealogical Society of Utah. In his address, “*Connecting Your Pedigree into Royal, Noble and Medieval Families” (3),* he described how reproduction between two individuals who share a common ancestor causes the number of distinct ancestors in the family tree of their offspring to be smaller than it would otherwise be.

To help conceptualize this, think of first cousins who share a set of common grandparents. If they were to produce a child, their child would **not** have the typical eight great grandparents but instead would have only six distinct individuals on that generational tier occupying the eight biologically necessary spaces on their heredity chart. To continue the illustration, unions between second cousins result in only 14 great-great grandparents instead of the usual 16. Third cousins have just 30 out of 32 GGG grandparents. The further apart the cousins the longer back the reduction occurs and the less severe it is as a ratio of necessary slots versus actual distinct individuals occupying those positions.

If our conceptualization of this phenomenon is not clear at this point please go back and read the last couple paragraphs again until we’re sure we’ve got it because the entirety of this thesis depends on a firm grasp of both this concept and its SOLE AND EXCLUSIVE ROLE as an exponential ancestorial expansion reducer. The relentless demands of biological law, rational logic and objective mathematics on the size of causally necessary progenitor populations for any extant human living today can only be checked in this one single way.

This now begs the question; can real world Pedigree Collapse do the job of reconciling the alleged populations of antiquity with the biological mandates and constraints of human propagation over even relatively short historical time frames. The remainder of this publication will demonstrate clearly that it cannot and that even the most preposterous claims of historical human inbreeding still fall orders of magnitude short of doing so.

Understanding the Landscape

The academic community has long relied on Pedigree Collapse as the explanation for resolving the ancestral paradox. However, this reliance represents both a misunderstanding of the problem's scope and a misapplication of theoretical research. To properly evaluate pedigree collapse as a solution, we must first clarify what constitutes genuine pedigree collapse versus related phenomena and examine how various demographic factors have been conflated in the literature.

Pedigree Collapse as the Sole Rational Mechanism

First, we all must understand that Pedigree Collapse represents the **only** mathematical and rational mechanism for reducing an individual's necessary ancestral populations against the LRGD. There is no way to fill all the biologically mandatory genealogical tree positions with fewer unique individuals than there are slots to fill than by any other means than Pedigree Collapse. Additionally, all proposed explanations for resolving the ancestral paradox fall into two distinct categories: direct applications of Pedigree Collapse, or circumstantial factors that influence its rate.

Category 1: Variations of Pedigree Collapse

Many researchers cite endogamy, polygamy, and related mating patterns as separate solutions to the ancestral paradox. However, these phenomena are simply different expressions of the same underlying mechanism - the appearance of identical individuals in multiple positions within an individual's family tree. Whether achieved through cousin marriage (endogamy), multiple wives sharing common husbandry (polygamy), or other patterns of concentrated incestuous mating within population pools, these practices all produce the same result; Pedigree Collapse. The terminologies and behaviors may vary, but the mathematical effect remains identical: reducing the actual number of unique ancestors below the theoretical 2^n predicted by the LRGD. This by definition, is what is meant whenever the term "Pedigree Collapse" is invoked.

Category 2: Drivers of Pedigree Collapse

Geographic isolation, population bottlenecks, and migration patterns are often cited when trying to hand wave Pedigree Collapse as a panacea for population paradoxes, but they do not constitute pedigree collapse but rather represent circumstantial factors that increase or decrease the likelihood of cousin marriages and related phenomena. These demographic conditions create environmental pressures that drive pedigree collapse rates higher or lower, but they are not solutions to the ancestral paradox in themselves, nor do they provide mathematical relief to the exponential doubling of the LRGD.

For example, geographic isolation increases the probability that individuals will mate with relatives simply due to limited partner options, thereby increasing pedigree collapse rates. Conversely, large-scale migration and population mixing tends to decrease cousin marriage rates, reducing pedigree collapse. These factors are important for understanding the variation in collapse rates across different populations and time periods, but they do not represent independent mathematical solutions to the exponential growth problem.

III THE ROBIN FOX CITATION CONSPIRACY

Historical Origins and Academic Folklore

Pedigree Collapse as a solution to demographic numerical paradoxes is built up as a cathedral of lazy genealogical orthodoxy—a doctrine delivered from academic pulpits as if it were revelatory truth. For decades, students and lay researchers alike have been browbeaten by the same ritual: if you dare question the impossible exponential march of

ancestors—64 in six generations, 1,024 in ten—some high priest will condescend and intone that Pedigree Collapse explains it all. No paradox, no magic, just collapsed branches on the family tree, so move along, nothing to see here.

But this narrative is not a triumph of science; it is a monument to intellectual hubris and institutional groupthink. The myth endures not because it is a proven fact, but because it is policed by a cadre of elite insiders—anthropologists, demographers, statisticians—who insist that their explanation is beyond reproach, that any public dissent is fueled by ignorance, and that alternative inquiries are the province of cranks.

Our thesis is not an exercise in gentle academic discourse. It is and unfortunately must be an autopsy. We will drag the pedigree collapse myth into the light; not just exposing its errors, but revealing the network of peer review collusion, citation laundering, authority transfer, academic hand waving and intellectual dishonesty that keeps it alive. The reader will see not only the cracks in the edifice, but the desperate hands patching over them—hoping, with each layer of deception and distraction to ward off scrutiny a little longer.

The mathematical impossibility of pedigree collapse theory to reconcile paradoxical populations of antiquity has been protected through systematic academic malpractice spanning at least four decades. Though we will address multiple failures of both academia and the concept itself, none rise to the level of one particular anthropologist. The case of Robin Fox possibly represents one of the most pernicious citation fraud conspiracies in modern scholarly history involving apparent coordinated deception, peer review manipulation, and likely institutional corruption.

While Robert Gunderson's original formulation was intellectually sound as far as it goes, the concept's subsequent evolution in academic and popular literature has been marked by problematic claims and unverified assertions. Most notably, the field has embraced a statistic claiming that ***"80% of all marriages throughout history have been between second cousins and closer"*** (4)(6)(7)(9) a chimerical statistic that appears to have originated not from rigorous research, but from a 1985 New Yorker Magazine article by Alex Shoumatoff, "*The Mountain of Names*", New Yorker Magazine May 13, 1985 (page 51).(4)

The Timeline and Exposure of a Citation Myth

Exhibit A:
Alex Shoumatoff's 1985 New Yorker Article

The origin of the famed *"80% of all marriages in history are between second cousins or closer"* (4)(6)(7)(9) claim traces not to Robin Fox's peer-reviewed research, but to Alex Shoumatoff's May 13, 1985, New Yorker piece "*The Mountain of Names*." Here, Shoumatoff writes: "*Robin Fox told me: 'If we could all get into the mind of God; we would know that 80% of all marriages in history are between second cousins and closer*.'" (4) This reporter's quote of Robin Fox represents this outrageous claim's debut.

Key points:

This is a journalistic anecdote, not primary research. The claim is hedged with the qualifying phrase "*If we could all get into the mind of God*"(4)—clearly indicating speculation or hyperbole, not an empirically established fact.

Exhibit B:
The 25-Year Academic Silence

For 25 years after Alex Shoumatoff's 1985 New Yorker article, Robin Fox never claimed the "80% cousin marriage" figure in any of his academic or peer-reviewed publications. Despite extensive scholarly output during this period, Fox never formally presented, defended, or restated this number in peer-reviewed literature. This prolonged silence is not an oversight; it reveals that the "80%" figure was never grounded in evidence Fox nor his peers deemed fit for proper academic presentation.

Exhibit C:
The 2011 Firewall and Escalation —
Tribal Imagination, Civilization and the Savage Mind's 90% Assertion

In 2011, Fox finally published a cousin marriage statistic himself—but not in a peerreviewed academic venue. Instead, he introduced it in his non-peer-reviewed book *Tribal Imagination*, *Civilization and the Savage Mind* (5). On page 68, Fox escalates the figure and repeats the important qualifier: "*If we could all get into the mind of God, we would know that probably* ***90*** *percent of all marriages in history have been between second cousins or closer." (5)* (emphasis added).

Fox furthermore goes on to explicitly explain his academic approach. On page 13 of this same book he states, "*I have eschewed the twin academic curses of nitpicky footnotes and padded bibliographies."* (5) This statement signals hubristic elitism and disdain for the norms of academic rigor. Fox situates his speculative statistic in a lengthy, densely printed work that holds academic protocols in contempt.

Exhibit D:
The Lewis Misquotation and Firewall

Less than a year after *Tribal Imagination's* publication, author Stephen Lewis cited exactly this same book in his academic article, *"How Many Ancestors Do You Have?" ,* The Wild Peak, March 4,2012, *thewildpeak.wordpress.com/20123/03/04* (6) but used the more marketable "80%" figure, NOT the updated "90%" version that Fox actually published. Lewis furthermore circumvented protocol and omitted the page number when citing this nearly 500-page volume.

This selective misquotation and lack of page number reference appear not accidental but rather a textbook case of citation laundering designed to insert the 80% figure in academic and public discourse. This action shielded the more inflated (and even more absurd) 90% claim behind a dense, hard-to-discover citation text. Making verification difficult but for the most diligent of researchers. No corrections or clarifications have ever been issued about this discrepancy.

Exhibit E:
The PNAS Scientific Journal Citation Fraud
Peer Review Complicity

The academic entrenchment of this pillar of Pedigree Collapse's false authority is finally and dishonestly executed in a key article published in the Proceedings of the National Academy of Sciences (PNAS) a journal with considerable prestige and rigorous peer review standards. This newer additional article, [*Chagnon, N.A., R.F. Shenk, M.K., Hames, R., & Finn, M. V. (2017) Cross-cousin marriage among the Yanomamo…Proceedings of the National Academy of Sciences (PNAS) 114(13), E2590E2597* *https://doi.org/10.1073/pnas.1618655114* (7) was written by Napoleon Chagnon and colleagues. This paper became a vector for legitimizing the insertion into

the lexicon of Robin Fox's ludicrous claim that 80% of all the world's marriages occurred between second cousins or closer, and as we'll see, this claim has become a central core tenet of the Pedigree Collapse cult.

Critically, Robin Fox himself served as peer reviewer for this article. Despite his direct involvement, the paper's appendix like supplementary materials included the use of Fox's own, "*80% cousin marriage*" figure without the crucial "*If we could all get into the mind of God"* qualifier, thus presenting this fabricated anecdotal speculation as empirical fact.

Within this context, the statistic is "incorrectly" attributed via citation to, A.H. Bittles, (7) whose work is referenced as the source for this "statistic" in the supplemental notes reference section. The total absence of any reference whatsoever to Fox's quote in, *Tribal Imagination,* (where the modified version of the statistic actually appears) anywhere in the entire reference section rules out any doubt that the wrong citation listing was simply a clerical error. The citation reads exactly as follows:

(79) "Bittles, A.H., 'The Prevalence and Impact of Consanguineous Marriage in Contemporary Societies,' Journal of Biosocial Science, 33(4), 2001, pp. 543-554."(7)(8)

This citation number 79, links to the following excerpt from Bittles et al*:*

"There is little information on inbreeding during the critical early years of human existence. However, given the small founding group sizes and restricted mate choices it seems inevitable that intrafamilial reproduction occurred and the resultant levels of inbreeding would have been substantial. Currently, couples related as second cousins or closer (F≥ 0.0156) and their progeny account for an estimated 10.4% of the global population. The highest rates of consanguineous marriage occur in north and sub-Saharan Africa…"(7)(8)
The Reference Fraud

What Bittles actually wrote is about contemporary data showing that 10.4% of the current global population are comprised of both the parents and offspring of cousin unions. What Chagnon's PNAS paper claimed: "*80% of all marriages throughout human prehistory were between close cousins*."(7) This 80% claim, in all of the contexts it's ever been used in is unsubstantiated, highly speculative and definitely a non-contemporary historical assertion as opposed to the A.H. Bittles quoted citation.

This misattribution represents the following multiple levels of fraud.
1. Magnitude Fraud: 800% value increase
2. Category Fraud: Children and parent statistics substituted for marriage statistics.
3. Attribution Fraud: Fox's unverified and speculative claim falsely linked to Bittles 4. Temporal Fraud: Contemporary statistic versus historical

The Smoking Gun:
Fox as Peer Reviewer

Robin Fox served as the lead peer reviewer for this article. This means Fox approved the use of his own 80% statistic. He most certainly knew the claim was his that he'd been informally making, (and via the cooperation of others), salting into the lexicon for three decades at this point. There's no way peer reviewer Fox didn't see the claim or the false attribution reference. It's the most fundamental and primary responsibility of the peer reviewer position. Fox approved the false attribution to Bittles for a claim Bittles never made and to which the documented citation did ***not*** apply.

This institutional malpractice enabled citation laundering that transformed speculative remarks into presumed fact. Obscuring the complete lack of evidence for the most major and most often cited factoid regarding pedigree collapse in literature that prescribes it as the end all cure all of exponential ancestor requirements. This make-believe claim continues to be falsely perpetuated under the guise of peer-reviewed science. Had one of Fox's undergrads done this they would have been put on academic probation or possibly expelled. But all in a days work for the "imminent kinship scholar". No correction or retraction has been issued regarding this misrepresentation

Internet Proliferation and Public Discovery (2017–2025) The Mythology Goes Viral

Following the PNAS legitimization, the 80% statistic exploded across:

1. Academic papers citing the fraudulent PNAS source
2. Blogs and popular science articles
3. Reddit discussions and online forums
4. Social media platforms

Independent Confirmation
The Reddit User Investigation

A comprehensive, independent investigation by an anonymous Reddit user provides powerful external confirmation of Fox's habitual fabrication of "80%" statistics.

https://www.reddit.com/r/wikipedia/comments/ds52t/80_of_all_marriages_in_history_have_been_between/(9)

"So, I have exhausted my research skills and I cannot find the actual data or research or original publication of this quote. I have searched nearly everything regarding Robin Fox, the anthropologist responsible for this statistic, but I can find nothing he has written to support it, or even where it comes from. Only lots of people quoting this. Can anyone provide the actual source of this statistic?

Update: In fact, after looking over a bit of his research, it seems '80%' is just the number he pulls out of his ass when he wants to make up statistics. Consider the following other revelations by the illustrious Robin Fox:
'It also distresses me that 80% of my colleagues act as if scholarly decisions can be made by legislation and resolution...'
'I am sure that over 80% of AAA members who voted to endorse the Seville Statement on Violence did so out of a sincere belief...'
*Yeah, I think 80% is Robin Fox's way of saying 'I have no f****** clue.' This should NOT be on Wikipedia." (9)*

Proliferation of the "80% Cousin Marriage" Statistic in Defense of Pedigree Collapse Theory

Following the complete and final legitimization of the "80% cousin marriage" statistic in the 2017 peer-reviewed PNAS paper, this statistic has become a central pillar actively invoked to defend the sufficiency of pedigree collapse theory as an explanatory resolution to genealogical paradoxes arising from retrograde doubling. Below are instances and citations from sources where the "80%" figure is cited explicitly with pedigree collapse discussed as the core resolution for paradoxical ancestor numbers.

The "*80% of all marriages being between second cousins or closer*" claim is indeed pervasive across academic, popular, and online sources, despite its complete lack of empirical grounding. Here is a summary of many instances where the statistic appears, illustrating how widely and deeply it has infiltrated discourse—always without verification, (because none exists).

1. Reddit discussion (2019): An exhaustive search found no original data or published source by Robin Fox supporting the 80% figure. The claim circulates widely but remains unsubstantiated, with critics calling it “just a number he pulls out of his ass” used as a rhetorical placeholder rather than fact.

2. NeoGAF forum (2016): Cites Fox as a Rutgers anthropology professor attributing the 80% figure to historical proximity and limited mobility, presenting it as common knowledge without source critique.

3. ZME Science (2023): Repeats the exact claim tied to Robin Fox, linking it to cousin marriage prevalence today but conflating contemporary and historical data.

4. Quarks Daily (2010): Calls Fox “the eminent kinship scholar” who “estimates fully 80%” of historical marriages are between first or second cousins, again without evidence.

5. FamilyTree.com (2017): Publishes the same statistic with a narrative about family locality patterns and cites Fox as source, again in defense of pedigree collapse’

6. Discover Magazine: Reports the 80% claim coming from Fox in the context of genetic counseling and risks associated with cousin marriage.

7. Wikipedia (multiple years): States “one estimate is that 80% of all marriages have been between second cousins or closer” with no direct source or deconstruction of the claim.

8. Various blogs and forums (including Familypedia) continue to echo the claim attributing it to Fox, demonstrating now its unquestioned status in internet discourse.

Contemporary demographic data

By contrast, careful modern research shows only about 10.4% of the global population today are the parents and product of unions between second cousins or closer—this includes both couples and their children (9)—underscoring the huge inflation and misapplication of the “80%” figure in pedigree collapse literature. The sheer volume and variety of these citations, across disciplines and platforms, attest to the phrase’s viral

spread. Yet nowhere is there empirical evidence for its historical truth or Fox's original data. The statistic functions as a mythos—an unsupported foundation that nonetheless propels pedigree collapse theory

The Deep Infection of the Discourse Ecosystem Including Artificial Intelligence

The legitimization of the "80% cousin marriage" figure in the 2017 PNAS paper catalyzed a viral proliferation that has deeply infected the entire pedigree collapse discourse ecosystem. This fraudulent statistic, paired with the fundamentally untenable and logically insufficient pedigree collapse theory, now functions as the default explanation deployed against challenges to retrograde doubling paradoxes.

Artificial intelligence knowledge platforms, trained on vast academic and public text corpora infused with this misinformation, routinely repeat the "80% cousin marriage" claim and uncritically present pedigree collapse theory as an adequate solution. Despite internal recognition within AI systems that pedigree collapse theory is inadequate or incomplete, the entrenched presence of the claim leads AI to deploy it as a primary answer to genealogical paradox queries, as well as dismiss alternative or critical perspectives as fringe or unsound.

This pattern represents a self-reinforcing cycle of misinformation, in which fraudulent statistics are cited and amplified across disciplines and popular media. AI systems and digital knowledge engines internalize and replicate the myth. The "infected" explanation crowds out genuine, rigorous inquiry and debate.

The myth's ubiquity and institutional backing have created an intractable misinformation ecosystem that defies correction, rendering it almost impossible for ordinary people or researchers to access accurate alternative explanations for retrograde doubling and related genealogical anomalies.

Full Academic Corruption and Its Lasting Impact: The 2017 PNAS Legitimization as the Linchpin

The inclusion of the "80% cousin marriage" statistic in the 2017 PNAS peer-reviewed article did not represent a tentative or partial acceptance of the claim—it marked its complete and final legitimization within the highest echelon of academic authority. This was no innocent error or minor oversight. Rather, it was the culmination of a decadeslong strategy of academic malpractice and citation fraud, executed at the highest professional

level by Robin Fox himself, who served as peer reviewer and actively enabled the deception.

This level of peer review corruption is the most egregious form of academic malpractice possible. A professor publicly advanced a fabricated statistic, weaponized his authoritative position to shield and smuggle it into peer-reviewed literature and did so using collusion, false and untraceable citations to obscure verification, and effectively cemented an unverifiable fiction into “scientific consensus.”

The PNAS paper provided the critical “academic anchor” that transformed journalistic speculation and non-peer-reviewed conjecture into seemingly incontestable scientific fact. This legitimization unleashed a tidal wave of amplification: academic papers, popular science articles, blogs, online forums, and social media began repeating the “80%” figure unquestioningly, embedding it deeply into the pedigree collapse discourse.

Most alarmingly, this entrenchment has spread into artificial intelligence knowledge engines. AI systems trained on the existing academic and public corpus—including digital encyclopedias, research databases, and curated knowledge bases—now routinely cite pedigree collapse and Fox’s “80% cousin marriage” statistic as the default explanation for genealogical paradoxes such as retrograde doubling. Despite internal algorithmic recognition that pedigree collapse theory is incomplete and methodologically inadequate, AI replicates the entrenched myth due to its overwhelming dominance in authoritative sources.

This phenomenon exemplifies a self-reinforcing cycle of cytogenesis, where a fabricated claim is inserted, legitimized, amplified, and then perpetuated by authoritative reference loops—including by cutting-edge AI—serving as de facto confirmation for the uninformed public, academics, and even advanced analytical systems alike.

In practical terms, when ordinary individuals inquire about genealogical paradoxes, the dominant, authoritative answer remains Robin Fox’s now fully but falsely legitimized “80% cousin marriage” figure as evidence of pedigree collapse theory efficacy. This reveals how academic gatekeeping and corruption have manufactured a durable myth that smothers genuine inquiry and critical examination.

The saga of the Robin Fox citation conspiracy ultimately lays bare how systemic academic malpractice—deeply ingrained in peer review and citation practices—has weaponized academic prestige to protect mathematically impossible genealogical claims through fraudulent means. It also reveals how such corruption can infect entire fields of study and public knowledge, creating a misleading intellectual orthodoxy resistant to

challenge or correction. Thus, when Alex Shoumatoff quoted Robin Fox quoting God, the mythos of Pedigree Collapse was born.

IV

THE SYSTEMATIC MISAPPROPRIATION OF CHANG STUDIES

Before we embark on dissecting the errant mathematics of Collapse theory, we need to undermine its flawed academic foundation a bit more. While Fox's claim stands apart for its dominance in the pseudo-scientific Collapse discourse, the work of Joesph Chang et. al., (10)(11) is misused nearly as much, albeit in an entirely different way.

The widespread acceptance of pedigree collapse theory as a resolution to the ancestral paradox depends not on rigorous mathematical validation but on the systematic misappropriation of prestigious academic studies that never addressed pedigree collapse as a genealogical mechanism. Two seminal works by Joseph Chang and collaborators—the 1999 paper "*Recent common ancestors of all present-day individuals*" (10) and the 2004 Nature study by Rohde, Olson, and Chang (11)—have been widely cited by proponents of pedigree collapse theory to lend unwarranted mathematical authority to claims they do not support.

This section examines both the nature of these studies and how their authority has been consistently and erroneously transferred, resulting in a pervasive citation abuse within genealogical literature and public discourse.

The Methodological Scope of the Chang Studies

Chang (1999): Random-Mating Models of Common Ancestry

Joseph Chang's 1999 study models genealogical ancestry through a carefully constructed mathematical framework grounded in coalescent theory and advanced probability. Using the Wright-Fisher model and Markov chains, Chang focused on *when* populations shared common ancestors in recent history, not how many of those ancestors there were. The objective was to determine temporal points of genealogical connectivity (most recent common ancestors, or MRCAs), not to count or model the unique number of ancestors per individual at each generational tier. (10)

Rohde, Olson, & Chang (2004): Simulation of Global Recent Common Ancestry

The 2004 Nature paper extends Chang's theoretical framework with computational simulations modeling recent common ancestry across worldwide human populations. Like the original study, it treats ancestry in a randomized, binary-lineage-per-child manner, enabling inference on population-wide genealogical connectivity timing. This approach again *does not* reconstruct real pedigrees nor quantify overlapping ancestors in the binary ancestral trees necessary to analyze pedigree collapse effects on individual ancestor counts. (11)

The Chang and Rhode Olsen studies both actually examine the exact opposite or reverse view of human propagation. The LRGD examines the explosion of genealogical ancestors back thru time STARTING FROM any one single person alive today. While the Chang and Rhode studies theorized how far back one single person propagated forward into all people living today, as well as to how far back theoretically every person from antiquity who have living descendants are collectively the same cohorts for any individual living today.

Neither of these studies speculate or even suggest the number of people in the target generations or ultimately how big the group of contributing cohorts may have been, nor do either study even make mention of Pedigree Collapse whatsoever. The authors never

explicitly or implicitly suggest or offer their work as justification for pedigree collapse as a paradox solution, nor do they even speak about the exponential genealogical growth requirements of binary reproduction.

Systemic Misappropriation Evidenced Through Multiple Examples

The inappropriate use of Chang's studies to substantiate pedigree collapse theory is no isolated occurrence but rather a systemic pattern observable across genealogical education, scientific literature, and popular media.

Example 1: Family Tree Magazine Family Tree Magazine asserts that Chang (1999) demonstrated mathematically how modern Europeans share recent common ancestors, further claiming this validates pedigree collapse as the mechanism controlling exponential ancestor growth. The magazine states: *"In a 1999 paper, Yale University statistician Joseph Chang used a mathematical model to show how all modern Europeans... have a common ancestor from about 1400. Pedigree collapse, the result of cousins marrying, is what prevents the number of ancestors from reaching astronomical numbers."* First, the one does not follow from the other. Secondly, this conflation misrepresents Chang's focus on genealogical connectivity timing as direct evidence of pedigree collapse effects on ancestor counts, which Chang's model does not address.

Example 2: International Society of Genetic Genealogy (ISOGG) Wiki the ISOGG's "Pedigree Collapse" article cites both Chang (1999) and Rohde et al. (2004) to explain why pedigrees must collapse: *"These models demonstrate how rapidly shared common ancestors can appear and why pedigree collapse is inevitable."* ISOGG presents the studies as validating and quantifying pedigree collapse (which is an actual phenomenon just not in the way pseudo-scientific people use it), despite the completely different research questions and modeling choices in the original works, thus lending unwarranted academic authority to unsupported claims.

Example 3: Lachance (2009), Peer-Reviewed Literature Joseph Lachance's peerreviewed article directly applies Chang's and Rohde et al.'s MRCA methodologies to estimate pedigree size and effects of inbreeding, stating: *"The number of ancestors in the pedigree of a single present-day individual can be calculated using biparental coalescence models (Chang, 1999), yielding estimates for global coalescence times."* While peer-reviewed, this work exemplifies the problematic extension of a MRCA framework into pedigree collapse calculations that require a binary pedigree structure and counting the numerical impact of consanguinity; something neither Chang model does. The underlying mathematical assumptions in Chang (1999) do not support such quantification of ancestor count reduction, making Lachance's conclusions susceptible to the same fundamental authority transfer misapplication.

Implications for Pedigree Collapse Theory and Academic Discourse

This consistent mischaracterization and reuse of Chang's authority highlights a systemic citation malpractice, whereby complex, mathematically rich studies are repurposed to legitimize pedigree collapse theory despite their fundamental inapplicability. This misuse obscures the genuine mathematical complexities of pedigree collapse. They also provide a false sense of resolution to the exponential ancestor paradox.

This is a pernicious habit amongst Collapse advocates, and it discourages rigorous development of accurate mathematical models incorporating empirical pedigree models and documented historical inbreeding rates. It undermines academic integrity by perpetuating a misleading narrative accepted in both popular and scientific communities.

Conclusion

Neither Chang (1999) nor Rohde, Olson, & Chang (2004) address or validate pedigree collapse mechanisms. Their exclusive focus on theoretical models designed for genealogical connectivity means they cannot provide mathematical substantiation for the ancestral count reductions that pedigree collapse theory claims. The systemic appropriation of these works to support pedigree collapse is a clear academic authority transfer error that perpetuates fundamental misunderstandings in genealogical science. Correcting this misuse is necessary to advance accurate mathematical treatment of human ancestry and ancestral paradoxes.

V ONLY METHODOLOGICALLY SOUND RESEARCH: ROBERT C. GUNDERSON'S EMPIRICAL INTEGRITY

We will now take a break from discussing the academic deficiencies within Pedigree Collapse advocacy as a resolution for paradoxical population quotas. Having given just cause to distrust the establishment's party line we will now look at the concept itself to gauge its sufficiency to hold back the existential and exponential tide that biological causality necessitates.

All the intellectual integrity or lack thereof in the world wouldn't matter if the basic mathematics supporting this theory were sound, replicable and empirically demonstrable. It is because they are none of these things that the above scholarly subterfuge exists. If the mathematics validating the claims of Pedigree Collapse proponents were as plain and simple as that of those disproving it none of the smoke and mirrors would be necessary. There would be no need of the cloak and dagger stealthiness or hood-winkery if the prima facia numbers added up.

Amidst the academic fraud, misappropriation and mathematical impossibilities that characterize pedigree collapse literature, one researcher stands apart: Robert C. Gunderson, who coined the very term "Pedigree Collapse" and remains the only scholar to have approached the question with methodological integrity.

Gunderson's Revolutionary Empirical Approach

While the academic establishment built elaborate theoretical castles on foundations of fraudulent citations and misapplied mathematics, Gunderson took a radically different approach. He examined actual genealogical data from real human beings.

In his 1980 analysis, Gunderson studied genealogies tracing descent from Edward IV, analyzing individuals who should theoretically have 65,536 genealogical positions

(2^16) at the 16th generation. His empirical findings revealed approximately 23,000 unique ancestors*, (John Becker OGS Families vol 38 no 3 1999)* (*12) for various individuals alive at that time, (1980). Representing a real-world collapse rate of 12.5% per generation, with a cumulative 65% total reduction at generation 16.

This cumulative reduction may seem significant but in later sections we will demonstrate that cumulative aggregation rates do not and cannot keep pace with the geometric progression of the LGDR. We will show that by the time the cumulative effect reaches even 99.9999% total reductions of unique ancestors versus genealogical positions still leaves **trillions** of impossible to justify mandatory unique ancestors by generation.

The Devastating Honesty of Real Data

Gunderson's empirical approach exposes the intellectual bankruptcy of theoretical modeling that has dominated the field for decades. Rather than beginning with desired conclusions and manipulating models to achieve them, Gunderson simply counted actual ancestors in actual family trees.

The Result: Mathematical Devastation

Using Gunderson's empirically derived collapse rate - the only such empirical real-world statistically large sample measurement available to the independent researcher of this study - the mathematical requirements remain astronomically impossible: (utilizing 27.5year generational cycles). Please remember the following numbers fully take into account actual real world Collapse rates. In other words, these are POST COLLAPSE numbers. (all mathematical proofs will be demonstrated shortly)

- 30 generations (825 years ago): 16.9 million ancestors required
- 50 generations (1,375 years ago): 426 billion ancestors required
- 75 generations (2,062 years ago): 1.36×10^{17} ancestors required

Theoretical Maximum: Still Catastrophic Failure

We must understand Gunderson's data represents one of the highest theoretical maximums for pedigree collapse because his sample consisted of one of the most endogamous populations in human history. We will not allow ourselves to be tempted to manipulate the data with cherry picked examples or wild unsupported speculations. If our data is skewed it’s skewed against our argument, not in support of it. If any particular data pool could be used to prove Pedigree Collapse theory true it would be this one for reasons we’re about to explore. It is because data such as this is so

devastating to their position that our friends on the other side of the aisle judiciously avoid the use of empirical genealogical statistics when modeling their theory.

The counted person by person and compiled genealogies that give us a calculable collapse rate of 12.5% per generation come from a population field of nearly 100,000 total individuals, spanning 16 generations (starting subjects are gen zero), over a period of 440 years involving families from Europe and America who migrated in masse to settle in isolated wilderness. A data pool of this size and scope is inarguably a valid and statistically significant sample to say the least. The only thing that makes this an imperfect sample is that it is a sample population skewed in **favor** of Pedigree Collapse. It gives maximum advantage to the counter argument. Gunderson's data came from a population of Latter-Day-Saints that:

1. Had multi-generational polygamy (concentrated lineages with shared ancestry)
2. Extreme geographic isolation (Utah settlement with minimal immigration)
3. Common geographic origins (Mormon pioneers from shared New England communities)
4. Royal descent concentration (multiple lines tracing back to common English royal ancestors who intentionally practiced endogamy)
5. Exceptional record-keeping (the most comprehensive genealogical documentation repository in existence).

If one of the most meticulously and carefully documented, endogamous populations in history, with systematic polygamy, geographic isolation and royal endogamy still produces collapse rates that fail catastrophically (which we are about to show), then normal human populations make the mathematical impossibility even worse.

What Gunderson Actually Did:
- Used real genealogical data from actual human beings
- Counted actual ancestors in documented family trees
- Reported findings honestly regardless of theoretical implications
- Established one of, if not the largest empirically derived collapse rates in academic literature (and which data set is shrewdly avoided because of its exculpatory nature)

Academic Mythology Approach:
- Relies on fraudulent citations (Robin Fox's unverifiable claims)
- Hijacks authority of irrelevant research
- Employs theoretical modeling without empirical foundation
- Manipulate models to achieve predetermined conclusions
- Ignore mathematical impossibilities through speculative hand-waving

Gunderson's work demonstrates what genuine scholarship looks like. He just reported the numbers and followed the evidence wherever it led. He was not invested in solving existential riddles or claiming to solve the whole of the problem. He was just incorporating some genealogical observations while instructing semi-lay people at a conference. Nowhere in his address did he claim to have the ultimate answer to the ultimate problem of exploding ancestral populations. His research only offers a partial mitigation to the overall incompatibility of math and logic with reality as we know it. Without intending to take sides his empirical findings destroy the pedigree collapse mythology as an ultimate solution for paradox equalizer more thoroughly than any theoretical critique could.

The Uncomfortable Truth

We will see that even this most favorable real-world data for pedigree collapse still results in mathematical impossibility within well-documented historical periods. Gunderson unwittingly provided the empirical proof that the concept he named cannot solve the problems it was **hijacked** to address. His intellectual honesty in reporting real data, regardless of theoretical implications, stands as a monument to genuine scientific integrity in a field corrupted by academic fraud and mathematical irrationality. The final irony is the only methodologically sound research on pedigree collapse is the same research that proves it cannot work.

The Inevitability of Mathematical Impossibility

Even if researchers have somehow underestimated the true effects of pedigree collapse, the exponential nature of LRGD ensures that any delay is temporary. The mathematical impossibility is not prevented but merely postponed by a brief period, no more than three or four hundred years at most. By 90 generations (450 BCE), the requirements of the LRGD exceed Earth's total mass. Pedigree collapse cannot stop the inevitable. It can only delay it by centuries, not millennia.

The temporal compression of the onset of this phenomenon reveals in stark relief the fundamental inadequacy of demographic solutions to the mathematical inevitability presented by the LRGD. No conceivable marriage pattern, migration rate, or population bottleneck can indefinitely resist exponential growth. The ancestral paradox emerges within well documented historical periods regardless of pedigree collapse assumptions.

The Core Mathematical Delusion Behind Academic Fraud

Having exposed the fraudulent academic foundations supporting pedigree collapse theory— from Robin Fox's systematic statistical fabrication to the coordinated citation

manipulation used to legitimize unverifiable claims— we now turn to examine the fundamental mathematical impossibility that these elaborate academic deceptions attempt to obscure.

The academic fraud exists precisely because the underlying mathematics cannot withstand honest scrutiny. No amount of sophisticated theoretical modeling, prestigious institutional endorsement, or peer review manipulation can change the basic mathematical relationship between linear and exponential functions.

The Fundamental Mathematical Delusion in Demographic Theory

For decades, multiple academic disciplines have engaged in increasingly elaborate theoretical contortions to explain how sporadic, limited linear reduction mechanisms can somehow overtake or reverse the relentless march of geometric progression. This represents perhaps the most sustained mathematical delusion in modern academic history—the persistent belief that linear subtraction can indefinitely constrain exponential multiplication.

The mathematical relationship between linear and exponential functions is not merely well established. It is among the most fundamental principles in all of mathematics. Any exponential function will eventually overwhelm any linear function, regardless of the linear function's initial magnitude or frequency of application.

The pedigree collapse hypothesis requires an even more mathematically absurd proposition: that sporadic isolated pockets of linear reductions can constrain continuous exponential growth. This would require 100 percent first cousin and even closer (sibling, half sibling, parent/child) procreation events to occur with:

1. **Perfect timing** to prevent exponential overflow
2. **Unnaturally large magnitude** to overtake geometric progression
3. **Sustained consistency** across multiple generations
4. **Universal coordination** across global populations

The Temporal Impossibility

The academic literature consistently ignores that even temporary suppression of exponential growth merely delays the inevitable mathematical explosion. If pedigree collapse reduces ancestral requirements by 90% at generation 30, this only postpones the mathematical impossibility by approximately 3.3 generations:

-

- **Without collapse:** 2^30 = 1.07 billion ancestors
- **With 90% collapse:** 2^30 × 0.1 = 107 million ancestors
- **Delayed explosion:** 2^33.3 = 1.07 billion ancestors

The exponential function resumes its inexorable growth, reaching the same impossible requirements within a trivial time delay. Considering this fact alongside the alleged 200,000 years (8000 generations) of human propagation on planet earth that must be accounted for regarding any extant human alive today, we come into drastic confrontation with impossibility. We must remind ourselves that no line of heredity may terminate in retrograde. If you're alive on earth today, you represent an unbroken genealogical line descended from the dawn of humanity. Your ascending line of ancestry does not and cannot fizzle out in 1000 or 2000 or even 100,000 years measuring back in time. ALL OF YOUR PARENTS HAD TO HAVE PARENTS ALL THE WAY BACK TO THE VERY FIRST HOMO SAPIENS, NO MATTER WHAT. Doubling the number of genealogically and biologically mandatory placeholders all the way.

The Scale Problem: Orders of Magnitude

Demographic theorists consistently underestimate the scales involved in exponential growth. By generation 50, ancestral requirements reach 2^50 = 1.13 × 10^15 (over one quadrillion genealogical tree positions). Even a 99.9999% cumulative pedigree collapse reduction still requires **10 billion** unique ancestors just 1375 years ago (compared to the alleged planetary population of 250 million at that time). Do we really imagine such a thing is possible? Crunch the numbers for yourself, we are fully assured and confident in every mathematical assertion made in the entirety of this publication.

The mathematical truth is no realistic linear reduction mechanism can address exponential growth that eventually will exceed the total atomic mass of Earth regardless of how aggressively we apply inbreeding modifiers over the course of 8000 generations. But we promised not to look beyond 2500 years.

The Logical Absurdity of Academic Persistence

Despite these mathematical certainties, academic disciplines continue elaborating theoretical frameworks that propose increasingly sophisticated mechanisms for the linear constraint of exponential functions:

- **Anthropology:** Cultural marriage patterns as exponential suppressors
- **Genetics:** Population bottlenecks as mathematical solutions
- **Demographics:** Migration patterns as growth limiters
- **Genealogy:** Record-keeping gaps as evidence for collapse adequacy

The Fundamental Error: Category Confusion

The academic persistence in defending linear solutions to exponential problems represents a fundamental category error. Scholars are attempting to solve mathematical impossibilities through historical speculation. No amount of anthropological theorizing about marriage customs can change the mathematical relationship between linear and exponential functions.

This is equivalent to proposing that cultural practices could override the laws of thermodynamics, or that migration patterns could suspend the principles of conservation of energy. Mathematical relationships are not subject to historical modification.

The Intellectual Bankruptcy Revealed

The sustained academic effort to explain exponential constraint through linear mechanisms reveals the intellectual bankruptcy of conventional demographic theory. When an entire field of study depends on mathematical impossibilities, that field has abandoned scientific rigor for ideological protection of predetermined conclusions. Multiple academic disciplines have spent decades elaborating sophisticated theories to explain how 2 x 2 might equal 3 under special historical circumstances.

Conclusion: Mathematical Honesty Required

The time for diplomatic academic discourse about "complex demographic interactions" has passed. The mathematical relationship between linear and exponential functions is not open to scholarly debate, cultural interpretation, or historical modification. The academic community's persistent defense of such theories represents either fundamental mathematical incompetence or willful intellectual dishonesty in service of protecting ideologically necessary conclusions from mathematical scrutiny.

*Gunderson counts the primary starting subject being gen 1 not gen zero. This is why John Becker cites Gunderson's data as covering 17 generations. The author of this study used the counting method of the primary starting subject's parents as gen 1 and the primary starting subject as gen zero. This is simply a counting nomenclature preference and has no impact whatsoever on the ultimate outcomes of genealogical math but may be confusing when checking the language choices in the relevant references. John Becker says, "17 generations" but for the sake of consistency of counting methods we have translated 17 into 16. This change again has no impact on the final mathematical outcome as long as translation across counting system remains consistent and accounted for which we have done.

VI MATHEMATICAL MODELING: TESTING PEDIGREE COLLAPSE SCENARIOS

To rigorously test the pedigree collapse hypothesis, we conducted three mathematical models with varying collapse rates, each beginning the collapse function at the 5th generation allowing for four generations of natural doubling which reflects the typical reality for nearly all persons throughout modern civilization living today. When applied to Gunderson's sample population this model results in an even higher than previously stated Collapse rate of 17% (vs the earlier 12.5%). Once again adding another counterbias disadvantageous to our theory's layer of argumentative insulation.

In accordance with reflecting reality in all our subsequent assertions and arguments we will operate with this higher 17% rate for the remainder of our study. This liberality to the opposition is just one more disadvantageous to our theory's concession to ensure no dishonest skewing of the numbers in our own favor, (as the opposition is wont to do) can possibly be claimed. And please remember what an incredibly high outlying collapse value this 17% figure represents when considering the population from which the sample was taken. At every juncture where bias could enter in we have premeditatively allowed the counter-bias to prevail. If we have cheated, we have done so AGAINST our models and theories.

Model Parameters

- Initial Generations (1-4): Natural doubling according to LRGD and in accordance with normative modern western Civilization
- Collapse Implementation: Starting with Generation 5 and beyond for EVERY generation we institute a reduction cascade by specified percentage per generation
- Generational Scope: Extended these calculations out to 25, 50, and 75 generations
- All values beyond the decimal rounded down in every calculation to eliminate partial human beings. This also serves conservatism contrary to our thesis's favor.

Model Formula

- $A(n) = 32 \times (2 \times (1 - r))^{(n-4)}$

- A(n) = Number of unique ancestors at generation n
- r = Collapse rate (expressed as decimal: i.e. 0.25 for 25% collapse)
- n = Generation (n>5)
- For generations 1-4 A(n) = 2^n

Model A: 15% Cascading Collapse Rate

Starting population at Generation 5: 32 unique ancestors

Generation	Pre-Collapse	Post collapse	Generation (n+1)
5	32	27	54
10	919	804	1608
25	4,063,956	3,555,961	7,111,922
50	1,275,852,184,993	1,084,474,357,244	2,168,948,714,489
75	6.15×10^{18}	5.22×10^{18}	1.05×10^{19}

Result:
With 15% collapse per generation, by generation 50 (1375 years back, 650BCE) we still require 1.08 **trillion** unique ancestors.

Model B: 25% Cascading Collapse Rate

Starting population at Generation 5: 32 unique ancestors

Generation	Pre-Collapse	Post collapse	Generation (n+1)
5	32	24	48
10	486	364	728
25	212,816	159,612	319,224
50	4,030,397,630	3,022,798,223	6,045,596,446
75	101,772,248,871,852	76,329,186,653,889	152,658,373,307,778

Result:
With 25% collapse per generation (significantly higher than Gunderson's empirical 17% rate and fully double the 12.5% rate), by generation 50, (650 C.E.) we still require 3.02 billion unique ancestors. At 75 generations, (2062 years ago, around the time of the birth of Herod) we must have 152.6 TRILLION. This, by the way, is the rate the fabled 80% second cousin or closer marriage statistic would produce if the alleged 80% cousin marriages were divided equally between first and second cousins, more on this later.

Model C 35% Cascading Collapse Rate
(more than twice Gunderson's empirical rate)

Starting population at Generation 5: 32 unique ancestors

Generation	Pre-Collapse	Post Collapse	Generation (n+1)
5	32	20	40
10	154	76	152
25	7,906	2,168	4,336
50	5,578,843	573,971	1,147943
75	3,936,660,767	2,558,829,498	5,117,658,99
100	2,777,869,246,556	1,805,615,010,262	3,611,230,020,524

Result:
Even with an extreme (and unrealistic) 35% collapse rate **per generation** we see only the temporary delay of inevitability with 2.5 billion unique ancestors just over 2000 years back for any one person living today. (nearly 2 trillion at 750 B.C. when Romulus was founding Rome)

Empirical Data from Gunderson

The most devastating evidence against the mythos of pedigree collapse comes from the only empirically derived large real world collapse rate known to the researchers of this paper. Gunderson reported that the genealogies of some of the descendants of Edward IV alive in 1980 show only approximately 23,000 unique ancestors at the 16th generational tier instead of the theoretical 65,536 (8,1), representing a continuous generational collapse rate of 17%, (starting the collapse cascade at gen 5)

This data is crucial because it represents the only real-world measurement being used by serious enquirers into the matter. All other studies use theoretical models loaded down with qualifiers and non-real-world assumptions. Let us take a look at a zeromodification measurement of the largest and most accurate genealogies known to exist. One of the highest rates possible,

Combination of Mormon and European royalty heritages.
Results Using the Empirical 17% Collapse Rate

30	16,902,862
50	4.27×10^{11}
75	1.36×10^{17}

Using the highest empirically documented collapse rate from one of the most endogamous large sample populations ever studied, we still require:

- **30 generations:** 16 million unique ancestors
- **50 generations:** 427 billion unique ancestors
- **75 generations:** 136 quadrillion unique ancestors

Exceptionally Favorable to Pedigree Collapse Theory Rates Still Inadequate by Orders of Magnitude

This genealogy represents an anecdotally high degree of collapse because:
- European royalty had mandatory endogamy
- Centuries of deliberate cousin marriages
- Abnormal endogamous cultural practices within royal circles
- Legal prohibitions on marrying commoners
- Multiple generations of LDS polygamous population

If even this selective "best-case" scenario still leaves quintillions of unique ancestors, then normal human populations with typical marriage patterns would require even more. If more than doubling this highly favorable rate still encounters paradoxical values within well recorded historical timeframes, then the time for new solutions is upon us.

Let us now return to dismantling the academic underpinnings of Pedigree Collapse's fallacious answer to impossible but biologically, mathematically and common sensibly necessary causality

VII CASE STUDY: HOW ACADEMIC PSEUDOSCIENCE PERPETUATES PEDIGREE COLLAPSE MYTHOLOGY

A Textbook Example of Academic Malpractice

To demonstrate how the pedigree collapse mythology is perpetuated through academic smoke and mirrors, we present a detailed analysis of a representative article. In *How Many Ancestors Do You Have*, Stephen Lewis, WildPeak@Wordpress 2012 (6), the author attempts to explain away the mathematical impossibility of human ancestry through Pedigree Collapse theory. This article serves as a textbook example of how pseudoscientific methodology and flawed mathematical reasoning combine to perpetuate demonstrably false demographic theories in scholarly literature. It's also the same article where Lewis and Fox gave the appearance of collusion to launder Fox's 80% claim.

Acknowledgment of the Problem
Then Magical Thinking

The article correctly identifies the exponential crisis: "*After thirty generations we would seem to have needed 1.1 billion direct and distinct ancestors! Yes, more than one thousand million direct ancestors at the time of William the Conqueror! Obviously, this is impossible*." (6)

Having acknowledged this mathematical impossibility, the author then proceeds to propose pedigree collapse as the solution. However, his own mathematical analysis will subsequently prove that even extreme pedigree collapse scenarios cannot resolve the problem he has identified.

The Devastating Inadequacy of the Author's Own Models

The article's mathematical modeling inadvertently proves the very impossibility it attempts to refute. The author models an extreme scenario where "*every single one of your ancestors married a second cousin". (6)* He then admits even this historically impossible rate still requires 4,356,616 distinct ancestors at 30 generations. He goes on to tell us this exceeds the entire estimated British population by a factor of more than 3:1. The author then concedes this scenario is "*extreme because historically the level of second cousin marriage was nowhere near as high*"(6). If 100% second cousin marriage (a rate never achieved in human history) still fails catastrophically, then realistic historical rates make the problem worse, not better.

Pseudoscientific Methodology and Citation Fraud

The Robin Fox Citation: "As an Aside" Evidence of Coordination

The article's most suspicious moment occurs with this seemingly casual insertion. "***Just as an aside**, according to anthropologist Professor Robin Fox of Rutgers University, in his 2011 book The Tribal Imagination: Civilization and the Savage Mind, it is in fact likely that 80% of all marriages in all of history have been between second cousins or closer.*"(6) (emphasis added)

The "*As an Aside*" Red Flag

The phrase "*as an aside*"(6) typically indicates both a lack of necessity and relevancy. When an author prefaces information with "*as an aside*," it means the information that follows is not necessary to their argument nor is it likely to be relevant - they're just saying it to say it. The pattern suggests the entire article may be academic theater designed to create a stage and platform to salt the lexicon with that particular statistic.

Empirical Refutation of the 80% Claim

The following quote is from John Becker, OGS FAMILIES, vol 38 no 3, 1999: "*A great deal is known about the family histories of the Amish who came to North America from Switzerland in the 18th century. It is estimated for one family about whom a very complete genealogy has been compiled that 21.5% of 627 marriages in this family were between 2nd cousins or closer*." (13)

A sampling of 627 marriages, (another statistically significant large empirical sample) from a religiously orthodox tightly knit community from the very era being suggested by

Fox and by extension, Lewis via citing it, to have cousin marriage rates of 80% being cited at 21.5% shows how untenable Fox's claim is. It also shows the radical difference between speculative theorizing and empirical investigation.

Theoretical Modeling Without Empirical Foundation

The Lewis article relies entirely on theoretical scenarios *("let's assume," "if we consider*") (6) without any reference to actual genealogical data or empirical evidence. This represents classic pseudoscientific methodology: beginning with desired conclusions and constructing theoretical justifications rather than examining evidence to determine what actually occurred.

Hand-Waving Complex Mathematical Relationships

The author promises to explain "*without using math's*" (6) what are fundamentally mathematical problems, then proceeds to misrepresent the mathematical relationships involved. Complex exponential functions cannot be adequately addressed through casual arithmetic examples that ignore cumulative effects and scaling relationships.

Fatal Conflation
Population Limits vs. Inbreeding

The article commits a fundamental logical error by conflating two completely different phenomena. "*But again, remember that the reversal of the pyramid to a diamond can only happen due to inbreeding, indeed being 'limited by population size' and 'inbreeding' amount to exactly the same thing*."(6)

Here, Lewis falsely claims population size limits and inbreeding are "*exactly the same thin*g."(6) On the contrary, they are in fact exactly not the same thing.

Inbreeding is an Observable and Quantifiable Demographic Phenomenon

1. Empirical genealogical data exists from centuries of documented family histories. Gunderson's research demonstrates measurable Pedigree Collapse driven by equally measurable INBREEDING occurred at a documented 12.5% rate for approximately 440 years (16 generations).
2. Amish community data shows actual empirically measured cousin marriage rates of 21.5% in what would be considered one of the highest endogamous documented populations, highly clannish Amish.

3. These documented real-world patterns show consistent human mating behavior across large and broad swaths of human demographic history
4. INBREEDING is a measurable demographic phenomenon that can be quantified via real-world genealogical trees

Population Size Limits
Subjective Model Container Estimates

1. Subjective assessments of what researchers believe local geographic and world populations were at various times
2. Speculative population estimates for ancient periods without reliable documentation 3. Model constraints used as containers to make mathematical models fit predetermined conclusions
4. Theoretical and presumed targets rather than empirical observations

The Circular Reasoning Problem

Lewis then resolves the mathematical impossibility by suggesting some black box mechanism that causes Pedigree Collapse to be more effective than he's already admitted it cannot be. This represents where the ***assuming the conclusion in the premise*** fallacy clearly takes place. Taking population estimates as axiomatic, then forcing models to fit those assumed containers regardless of empirical evidence.

The 1200 CE Event Horizon
Where Science Becomes Speculation

The critical transition point occurs around 1200 CE. It’s at this point that all pedigree collapse models encounter the same crisis. Genealogical data becomes sparse, or nonexistent making itemized head counts unavailable, but this is akin to a prosecutor withholding exculpatory evidence, then using the absence of proof of innocence as equivalent to proof of guilt. Fortunately, we may object.

Your Honor, the lack of empirical evidence gives way to theoretical speculation. Now all the models must suddenly invoke sudden dramatic changes in collapse rates (51%+ required for contraction). We must understand at this point the only way for Pedigree Collapse to explain demographic discrepancies is for the numbers of unique ancestors TO GROW SMALLER EACH GENERATION IN RETROGRADE when genealogy demands they not just grow larger but double.

Imagine an upside-down pyramid. You are the point at the bottom. Your genealogical tree is the ever-expanding triangle growing upwards. Now, Pedigree Collapse will elongate this shape and have some mitigating effect on its expansion. But once ancestorial counts reach full saturation of alleged temporal human population limits this expansion must not only stop, but REVERSE into a contraction effect (the “diamond” shape that’s never been observed in any real world LARGE data sample, but only speculated in models that don’t start the diamond shape contraction until they’re beyond the available data fire wall of 1200CE) if Pedigree Collapse theory as a solution to the paradox is to hold true. Each next successive retrograde generational tier must have fewer unique individuals in it than the preceding tier.

For example, however many unique ancestors you have in generation 29 you must have fewer unique ancestors in generation 30, even though every member of the gen 29 cohorts had to have two parents each. We concede that this far back the number will be significantly less than **double** the number of generation 29, but if that number is not reduced by at least 51% no contraction takes place. A 49% collapse rate results in a small amount of continuing expansion. If A is doubled into B, and then B is reduced by anything < 50%, B will still be larger than A and your number of ancestors by generational tier continues to grow. For B to be less than A, B must be reduced by >50%. Otherwise, the fabled “diamond” shape never takes place.

Remember, 35% fewer retrograde cohorts per generation **iterated for every generation** still results in unmanageable expansion that exceeds earth’s carrying capacity in less than 2500 years and represents twice any empirically derived observable rate ever measured. Models must invoke and speculate about impossible to verify sudden accelerated changes in collapse rates as soon as the irrefutable data to the contrary runs out. They cannot ever impose these unrealistic scenarios on more recent populations because the available records will immediately contradict them.

Inadequate Mechanisms for Impossible Outcomes

The article's central failure involves proposing a mechanism that cannot mathematically achieve the required results. Lewis acknowledges that unrealistically extreme close cousin marriage rates fail catastrophically, then proposes that the cumulative effects of distant cousin relationships will somehow create dramatic demographic transitions around 1200 CE without providing mathematical demonstration.

Lewis's proposed mechanism has the additional illusionary effect of being extremely difficult to falsify. Genealogical research into distant cousin relationships becomes extraordinarily complicated - most people could never figure out what a 15th cousin relationship represents, much less quantify their statistical impact across populations.

This creates a veil that shrouds statistical claims behind a firewall of academic expertise and complex mathematical skills. While an armchair researcher can easily verify first and second cousin mathematics, discovering actual distant cousin relationships in real genealogical data would require graduate-level statistical algorithms and extremely skilled data analysis.

The Modern Oracles of Delphi:

By invoking distant cousin relationships, pedigree collapse theorists have created an academic priesthood and become the modern Oracles of Delphi. Their claims become esoteric knowledge accessible only to specialized experts, effectively reducing the number of people capable of empirical verification. This creates an implicit appeal to specialized authority: "This is too complex to explain to you, so just take our word for it."

Like the ancient Oracle whose cryptic pronouncements couldn't be questioned by common people, these academic oracles deliver mysterious demographic mechanisms protected by computational complexity rather than empirical evidence. The mechanism becomes unfalsifiable not because it's scientifically sound, but because it's computationally inaccessible to most critics.

Lewis's "Must Be" Fallacy
The Methodological Breakdown

For ancestral numbers to contract (reverse direction), collapse rates must exceed 50% per generation. This requires close cousin marriage rates at 100% for multiple consecutive generations. Such extreme inbreeding would cause population collapse from genetic maladies. There are no historical precedents or documented evidence for such extreme endogamy. For the Pedigree Collapse advocates models to work they must invoke theoretical breeding patterns that after 1000 years of having maintained level consistency for 30+ generations to suddenly accelerate 300% without explanation. A stable 17% rate jumps to >50% just after the data that holds them accountable in more recent eras dries up and becomes veiled by historical record deficits.

Academic Ego and Condescending Pedagogy

The "Simple Explanation" Fallacy

The article adopts a condescending tone ("*You might have even heard statements*...," "*Obviously it's a lot, but how many*?") (6) that positions the author as patiently

explaining complex realities to confused readers. This pedagogical stance masks the author's fundamental failure to address the mathematical impossibilities they acknowledge.

False Complexity as Intellectual Cover

The author references veiled "*complex mathematics"* (6) and pseudo-sophisticated genealogical concepts to create an impression of deep expertise while simultaneously admitting they cannot actually solve the mathematical problems they're discussing. This represents classic academic ego protection - appearing knowledgeable while avoiding mathematical accountability.

The "Magic Transition" Fallacy

When confronted with mathematical impossibility, Lewis abandons empirical analysis:
- "*Must have*" replaces "*could have*" - speculation becomes assertion
- Models are forced to fit predetermined population containers
- Theoretical mechanisms are invoked without demonstration (unquantified multirelational couplings)
- Circular reasoning as conclusions are assumed in premises.

Lewis's Proposed Mechanism

Lewis says that multiple distant cousin relationships within the same couple (1st cousins who are also 5th cousins, 8th cousins, etc.) will create cumulative pedigree collapse effects that appear insignificant for 30 generations but suddenly become dramatically effective around generation 25-30.

The mathematical reality is that distant cousin relationships (5th cousins, 8th cousins,15th cousins) produce negligible Pedigree Collapse effects. Remember these types of relationships eliminate perhaps 2 or 3 ancestors 10-12 generations back. Losing two, four or even a dozen out of 1024 at ten generations back will not effectively stem the tide of exponential expansion. This is a mathematical delusion. These relationships would need to accumulate over 30 generations while showing no measurable effect, then suddenly produce 51%+ collapse rates precisely when genealogical data becomes scarce and estimated population constraints are encountered.

The Timing Problem

Lewis's mechanism requires that imperceptible cumulative effects coincidentally become dramatically effective at exactly the moment when genealogical documentation disappears, population saturation points are reached, and model constraints demand mathematical impossibilities. How convenient for them that all this works out when they control the number of multi-relational couplings there are in the model.

They do not follow an empirical data set when making their calculations. They create a theoretical data set of intermarried couples that PRODUCE A MATCHING INDIVIDUAL ANCESTOR VALUE EQUIVALENT TO/LESS THAN THE ASSUMED HISTORICAL POPULATION OF THE TARGET ERA. Then claim model efficacy and theory validation presenting their conclusions as factual historic breeding statistics. Stating it “must have” happened this way because the assumed historical population estimates are considered factual premises which they customize models to tailor fit and conclude that their starting premise and final determination match so they’re right!!! When in fact none of these things is established by such methodologies and would be given an “F” by any college professor from any other field of study. This represents premeditated mathematical manipulation rather than demographic discovery.

Direct Quote Analysis

“*Such a phenomenon is called 'pedigree collapse'... Essentially there are two interrelated reasons why the number of your direct ancestors does not keep on exploding and why their number will sooner or later start to contract*." (6)

The phrase "*will sooner or later start to contract*" represents pure assertion without mathematical justification. The author provides no mechanism (other than his “black box” of multi-relational couplings) capable of overcoming exponential growth yet confidently asserts contraction "will" occur. This reasoning is based on the assumed premise of population sizes that serve as targets or containers that they then tool their models to fit into, assuming their conclusion in their premise.

The Wachter Model: Academic Sleight of Hand Exposed The Sophisticated Deception Technique

Lewis's discussion of Kenneth Wachter's 1980 model represents one of the most sophisticated examples of academic sleight of hand in the Pedigree Collapse literature. Through careful analysis of Lewis's presentation, we can observe exactly how model artifacts are transformed into "empirical evidence" through linguistic manipulation.

Step 1: Establishing Academic Gravitas Through Faux Transparency

Lewis begins by citing Wachter's credentials while embedding extensive disclaimers about the assumptive nature of the work:

> "*At this point I would like to highlight one of the seminal articles on this subject, called Ancestors at the Norman Conquest, published in 1980 by Kenneth W Wachter, a Berkeley statistician and demographer. Wachter *****assumed***** that 'England' was a closed society, with basically no international immigration. He uses the example of a *****hypothetical*****, strictly English, person born in 1947 (let's please not open a debate on what 'English' means) and ***\

to accord quite well with my earlier observations regarding my own ancestry. ***But then something peculiar starts to happen****."(6)9*emphasis added)

Analysis

The phrase "*something peculiar starts to happen*" implies natural demographic processes are being observed rather than (artificial) model constraints and output being calculated.

Step 3: The Critical Sleight of Hand

Lewis continues with the key deception:

> "*Going back only five more generations (to Generation 20), or to about the year 1377, the number of distinct ancestors has grown enormously, to 628,576, but this is now 'only' 60% of the theoretical maximum of 1,048,576 – meaning that 40% of the theoretically available ancestor 'slots' are now duplicates or better said non-distinct. Also if the English population at the time was about 2.25 million as Wachter assumes (remember the Black Death decimated the English population starting at about this time) then this individual's ancestors in 1377 (and only in one generation) now account for fully 28% of the total population alive at the time*."

**Analysis: ** Lewis presents these figures as demographic observations while concealing that they are model outputs constrained by assumed population limits.

Step 4: The False Contraction "Evidence" –
Dropping All Disclaimers

The deception culminates when Lewis presents the critical statistics without referencing their assumptive theoretical origins:

> "*25 generations ago, in about 1227, this individual would seem to have 2,012,114 distinct ancestors: or 80% of the estimated English population of 2.5 million.* ***Not only that but only about one sixteenth of his/her ancestors are distinct people****, the rest are at least duplicates. Going back 30 generations to around 1077, to just after the Norman Conquest, we saw earlier that the maximum number of ancestors in that one generation was just over one billion (i.e. 1,000 million). Wachter's model calculates that* ***this individual would have*** *952,279 distinct ancestors in 1077 – only around 0.09% of the maximum but representing fully 86% of the total estimated English population of 1.1 million."(6)*

The Linguistic Deception

Notice how Lewis now presents these numbers without any reference to their assumptive theoretical origins. The extensive disclaimers from the methodology section have disappeared. A casual reader would interpret these as empirical demographic findings rather than outputs from a hypothetical model based on unquantified assumptions. Lewis can claim complete transparency because he initially disclosed the assumptive nature of Wachter's work. However, when presenting the crucial statistics that form his model outputs, he drops all qualifications and presents them as factual demographic data.

Step 5: The Fraudulent Conclusion
Syntactic Transformation

Lewis then draws his false conclusion using completely different syntax than employed in the methodology section:

> "*In terms of the theme of 'Pedigree Collapse', the important point to notice is that between 1227 (i.e. generation 25) and 1077 (generation 30**) the number of distinct ancestors more than halved**, from just over two million to just under one million – **and that means precisely and unequivocally that the 'pyramid' became a 'diamond'!** In his model the reversal point takes place somewhere around 1200. So even with an extremely modest level of historical cousin marriage this model clearly shows that such marriages of **relatives will first narrow the pyramid and then put it into reverse** – i.e. it **clearly demonstrates** the phenomenon of Pedigree Collapse.*"(6) (emphasis added)

The Syntactic Deception

Lewis employs dramatically different syntax when describing methodology versus conclusions:

Methodology Syntax

- Tentative, qualified, assumption-heavy
- assumed, hypothetical, assumptions (repeated 4 times) - Conditional and uncertain language

Conclusion Syntax

Definitive, declarative, empirically confident

- "*precisely and unequivocally", "clearly shows", "clearly demonstrates"*
- Absolute statements presented as factual discoveries
- No reference to the assumptive theoretical origins
- Should have used language like "could be", "If our assumptions are correct the following is a possibility", or better yet, "model parameters demonstrate that given sufficient amounts of theoretical inbreeding a diamond chart shape can be achieved, but at the present time no such sustained levels of inbreeding are known to have ever occurred."

This syntactic transformation makes theoretical model outputs appear as empirical realworld data. The casual reader experiences a psychological shift from understanding they're reading about assumptions to believing they're reading about demographic facts. Even we, the authors of this study concede that given enough tweaking valid math can produce cousin marriage statistics ON PAPER that will cause adequate to their theory's need to prove that multi-generational contraction is possible ***theoretically***. But this is the same approach to reality as Maurits Escher's two-dimensional drawings that represent impossible 3-dimensional objects, such as his famous ever ascending stairs that lead back to their origin at the bottom. We see here what's possible on paper versus what actually occurs in reality can often be two very different things.

What Actually Happened

The Truth: Wachter didn't "discover" that ancestor numbers naturally or actually did contract. He simply showed that a programmed model can be forced into contraction if the exponential growth has to fit within assumed English population limits of ~1.1 million people.

The Process

1. **Started with conclusion** - ancestors must fit within ~1.1 million English population
2. **Worked backwards** - adjusted model parameters to produce required result 3. **Presented output as discovery** - claimed the forcibly manipulated contraction proved that pedigree collapse theory was not just possible but works and actually ("must have") occurred as they say.

The Mathematical Reality

The contraction from 2 million to 950,000 ancestors over 5 generations represents an average reduction of over 51% per generation - a rate that would require first cousin marriage levels approaching 100% with additional uncle-niece and sibling relationships. No historical evidence supports such extreme endogamy in medieval England.

The Ethnocentric Bias

Lewis's analysis reveals another critical flaw:

> "*This is ultimately the basis for the assertion highlighted at the beginning of this essay that everyone with English ancestry today is descended from 86% of people alive in the country at the time of William the Conqueror."(6)*

The Problem: This entire theoretical framework focuses exclusively on English populations while ignoring the identical mathematical impossibilities faced by:
- Chinese populations over the same period - African populations over the same period
- Native American populations over the same period
- Every other human population in history

The Devastating Timeline Problem

The article focuses on 30 generations (~1066 CE) but completely ignores that the exponential expansion never stops pressing against whatever mysterious ephemeral phenomenon they suggest is able to resist it. At 40 generations, even extreme pedigree collapse rates require billions of ancestors. At 50 generations, the requirements reach trillions. The timeline to absolute mathematical impossibility makes the problem unsolvable regardless of cousin marriage assumptions - and this impossibility emerges within well-documented historical periods, not in some distant prehistoric past.

Unpacking Wachter's and Lewis's Black Box The Countermanding Exponential Function

The only way to overcome an exponential function is with an additional countermanding exponential function. Wachter and Lewis are claiming their black box contains a countermanding exponential counter-function that is able to overwhelm the exponential growth of ancestral requirements. This mysterious function begins contracting generational trees somewhere between generation 20-30, requiring collapse rates in excess of 50% to achieve actual numerical contraction.

The Mathematical Requirements Exposed

Using Wachter's own theoretically and methodologically unsound numbers from Lewis's presentation, the contraction from generation 25 to generation 30 requires a sustained 56.9% collapse rate. Let's give them the benefit of the doubt and assume this mysterious black box of "distant cousin cumulative effects" somehow explains how they force population figures to fit within historical population containers.

The Inertia Problem
From Solution to Extinction

The problem is that the inertia of their model very quickly and exponentially outstrips the exponential growth it was designed to counter. This is because their model CANNOT begin to impose massive Collapse rates while existing data will prove it never happened that way. They have an incredibly narrow window of opportunity in which to tame the exponential increases they seek to eliminate. They have just 5 generations to turn (by their own admission) over two million ancestors per generation into 950,000 and do so against the urgent demands of the LRDG.

But, here’s the problem, their black box mechanism does such a good job of keeping ancestral numbers within alleged historical population containers that if left in place it dwindles all ancestors down to zero within documented human history - specifically by 1528 BCE, during the height of ancient civilizations.

The Black Box Sleight of Hand

Once Wachter's study manages to forcibly compress genealogical propagation to fit within alleged historical population limits, he doesn't explain what happens next. Once target values are reached, he "repacks” the black box but does not clarify that the historical containment problem is more than just solved, its eliminated and made moot by the extinction of humanity altogether by 1528 BCE. If these mysterious forces can accelerate Pedigree Collapse to nearly 60 percent per generation and are the result of CUMLATIVE effects, then wouldn’t they continue ACCUMULATING?!? If the collapse rate is truly cumulative (as Lewis claims), then leaving it constant at 56.9% represents extraordinary generosity to Wachter's model. We haven't even begun to speculate what the reality of accelerating cumulative effects would dictate.

The Geometric vs. Geometric Battle

Wachter's model claims that logarithmic increases in collapse rates can overwhelm the geometric progression of ancestral requirements. Meaning his geometric progression must be stronger than natural demographic retrograde growth. Yet he provides no

mechanism for this counter-exponential function beyond vague assertions about “multirelational cumulative effects."

The Convenient Black Box Strategy
The devastating pattern emerges

1. **Unpack the black box** when exponential growth creates impossible ancestor requirements
2. **Apply mysterious mechanisms** (56.9% collapse rates with no historical evidence)
3. **Achieve the desired numerical results** (fit within population containers)
4. **Repack the black box** before the mechanism's logical consequences become apparent

The result is Wachter throttles his model to begin reversal at generation 25 with a 56.9% collapse rate, then mysteriously stops explaining what this rate produces when extended to its logical mathematical conclusion over the course of the next 20 or 30 more subsequent retrograde generations.

If Wachter's black box mechanism is scientifically valid, then:
- **It must work consistently** when applied beyond his chosen parameters - **It cannot be selectively activated** only when convenient for fitting population containers
- **It must account for its ultimate mathematical destination** (human extinction by 1528 BCE)

Wachter's black box works perfectly for solving his immediate problem (generation 2530 population constraints) but becomes mathematically catastrophic when extended to its logical conclusion. This suggests the mechanism was designed backward from the desired result rather than discovered through empirical analysis.

CATALOGUE OF LOGICAL FALLACIES IN LEWIS’S ARTICLE WITH CONCRETE EXAMPLES

Below, each named logical fallacy is presented together with a direct, concrete example from Stephen Lewis's article, followed by concise commentary that demonstrates how, where, and why each is committed.

1. **Appeal to Authority (Argumentum ad Verecundiam)**

Example:

"Just as an aside, according to ***anthropologist Professor Robin Fox of Rutgers University, in his 2011 book****, The Tribal Imagination: Civilization and the Savage Mind, it is in fact likely that 80% of all marriages in all of history have been between second cousins or closer."(6) (emphasis added)* **Commentary:**

Rather than supplying any primary data or **evidence** for the 80% claim, Lewis appeals to Robin Fox's academic stature and simply relays the figure as authoritative truth. No demonstration or empirical support for this assertion is ever provided.

2. **Circular Reasoning (Petitio Principii) Example:**

Lewis assumes that pedigree collapse must resolve the exponential ancestor paradox, and then uses models constructed with that goal to "demonstrate" that it does.

Commentary:

He begins by declaring that ancestor numbers "*obviously must have collapsed because the population was limited*," then builds mathematical scenarios that force this outcome, using the conclusion as a premise.

3. **Moving the Goalposts / Mathematical**

Impossibility

Admission Example:

"100% second cousin marriage is an extreme because historically the level of second cousin marriage was nowhere near as high." (6)

"If every one of your ancestors married a second cousin, at 30 generations you would still have 4,356,616 distinct ancestors."(6)

Commentary:

Upon admitting that even implausibly high cousin marriage rates cannot solve the exponential problem, Lewis shifts the discussion to more vague, hypothetical, or compounded collapse mechanisms, never addressing the original scale of impossibility.

4. **Straw Man**

Fallacy Example:

Lewis introduces irrelevant analogies about small, isolated populations (e.g., the Robinson Crusoe scenario) and animal pedigrees.

Commentary:

He diverts attention from the actual mathematics of human demography by attacking oversimplified or inapplicable models, sidestepping the real issue of exponential ancestral proliferation in normal populations.

5. **False**

Equivalence Example:

"Being 'limited by population size' and 'inbreeding' amount to exactly the same thing."

Commentary:

He equates an empirically measurable, documentable phenomenon (inbreeding) with an abstract, assumed mathematical constraint (population size), treating both as identical mechanisms when they are fundamentally different.

6. **Appeal to**

Complexity

(Argumentum ad Obscurum) Example:

"I will avoid using complex mathematics … but you will see why the numbers don't add up."

Commentary:

Lewis implies that the mathematical issues are so intricate they cannot be addressed within the article, while simultaneously making sweeping conclusions that require exactly that same degree of mathematical rigor.

7. **Special Pleading Example:**

Lewis dismisses historical impossibilities (such as needing 100% cousin marriage rates) by suggesting that "complex historical factors" or undocumented cumulative effects "*must have*" occurred.

Commentary:

He excuses critical empirical deficiencies and mathematical failure by invoking unnamed special circumstances, not evidence.

8. **Argument from Ignorance (Argumentum ad Ignorantiam) Example:**

Lewis repeatedly states that because precise genealogical records do not exist for earlier periods, speculative models must be accepted.

Commentary:

Lack of historical data is used to justify or validate claims that cannot be grounded in empirical fact.

9. **False Dilemma Example:**

Lewis frames the debate as a binary: "*Either you accept pedigree collapse or accept that the number of ancestors is impossible given historical population sizes*."

Commentary:

He ignores alternative explanations or the possibility that the underlying assumptions of both positions require scrutiny.

10. **Confirmation Bias Example:**

Lewis highlights any figures or language that support the feasibility of pedigree collapse yet downplays or ignores quantitative evidence (like his own model results) that actually disprove his thesis.

Commentary:

He selectively interprets or reports data to support a predetermined conclusion.

11. **Red Herring Example:**

Discussions of migration, founder effects, and animal pedigree management appear as sidetracks.

Commentary:

These topics, while tangentially related, do not address the primary mathematical problem at hand, distracting from the core issue.

12. **Hasty Generalization Example:**

Lewis extrapolates from rare or unusual cases (such as island or "lab-created" populations) to generalize about human demographic patterns over millennia.

Commentary:

He draws universal demographic conclusions from atypical, outlier scenarios.

Each example above is tied directly to explicit statements or scenarios in Lewis's article, making clear how these fallacies function and accumulate to undermine the article's intellectual credibility.

The "Genesis Proof" Parallel
Exposing the Methodological Bankruptcy

The article's methodology exposes a fundamental flaw that renders the entire approach academically worthless: The same theoretical modeling techniques could be used to "prove" any predetermined conclusion. Using identical procedures, one could easily construct a model that "demonstrates" biblical special creation with Adam and Eve as a literal founding couple 5,000 years ago.

The Methodological Parallel

Following the article's exact approach:

1. **Acknowledge the exponential problem** - accept that geometric progression creates impossible ancestor requirements
2. **Identify the "black box" transition point** - locate the historical period where evidence becomes scarce (around 1200 years ago)
3. **Apply theoretical throttling mechanisms** - invoke hypothetical demographic factors to force the progression into a "diamond shape"
4. **Reach the predetermined conclusion** - throttle the model down to exactly 2 people (Adam and Eve) at precisely 5,000 years ago
5. **Claim scientific validation** - declare that mathematical modeling has "proven" Genesis

The Identical Logical Structure

The Genesis model would employ the same rhetorical strategies as the pedigree collapse article:

- Complex theoretical mechanisms ("divine demographic optimization," "supernatural breeding constraints")

- Confident assertions without proof ("the evidence clearly shows population convergence toward a founding pair")
- Mathematical hand-waving ("while the calculations are complex, they demonstrate biblical accuracy")
- Appeal to authority ("leading researchers in theological demographics confirm...")
- Dismissal of impossibility concerns ("critics fail to understand the sophisticated dynamics involved")

The Devastating Implication

If the same methodological approach can "prove" both naturalistic shrinkage of ancestral populations and biblical special creation, then the methodology itself is fundamentally worthless. The article's techniques represent **conclusion-driven modeling** rather than evidence-based analysis. Any theoretical framework that can be manipulated to support contradictory predetermined conclusions has no scientific validity whatsoever.

Exposing Academic Bias

The acceptance of pedigree collapse "proofs" while rejecting Genesis "proofs" using identical methodology reveals the arbitrary nature of academic gatekeeping. Both conclusions rely on:

- Unverifiable historical assumptions
- Theoretical modeling without empirical validation
- Mathematical manipulation to reach desired endpoints
- Dismissal of mathematical impossibilities through speculative mechanisms

The only difference is which conclusion the academic establishment finds ideologically acceptable, not which conclusion the real-world empirical evidence actually supports.

The Ultimate Critique

This parallel demonstrates that the article's approach represents pure academic theater; sophisticated-sounding analysis designed to rationalize predetermined beliefs rather than determine what actually occurred. When identical methodologies can "prove"

mutually exclusive conclusions, the entire theoretical framework collapses into intellectual bankruptcy.

The pedigree collapse mythology persists not because of its mathematical validity (which is nonexistent) but because it provides academically acceptable explanations for ideologically unacceptable phenomena. The methodology's flexibility in supporting any desired conclusion exposes it as pseudoscience masquerading as scholarship.

The Collective Academic Delusion

The article analyzed here represents more than individual scholarly error - it exemplifies a **collective academic delusion** where an entire field continues to elaborate on, defend, and teach a theory that:

- **Fails its own mathematical tests** (their 100% cousin marriage model still requires impossible populations)
- **Relies on fraudulent citations** (the Robin Fox mythology)(Chang studies misappropriation)
- **Employs methodologically bankrupt approaches** (conclusion-driven modeling that could "prove" anything)
- **Ignores mathematical impossibility** (exponential functions that exceed Earth's atomic mass)

The Uncomfortable Truth

Human ancestry, as conventionally understood, is mathematically impossible under natural conditions. No amount of academic posturing, theoretical sophistication, or professional authority can change the arithmetic. The numbers don't work. They never worked. They can't work.

The Emperor Has No Clothes

Just as in the children’s fairy tale one simple honest observation shattered the collective delusion in Andersen's (14) story, simple childlike mathematical honesty must shatter

the Pedigree Collapse mythology. The time for diplomatic academic discourse about "population dynamics" and "complex demographic factors" has passed.

VIII

INSUFFICIENT CAUSES AND CAUSAL RUPTURE

THE END OF NATURALISTIC EXPLANATION

In the preceding chapters, we have pursued—we believe to the point of mathematical exhaustion—the constraints imposed by biological law and arithmetical necessity upon the development of human ancestry. We have shown, with unblinking fidelity to logic and evidence, that the relentless doubling demanded by binary sexual reproduction, as encoded in what we have termed the Law of Retrograde Generational Doubling (LRGD), creates a genealogical architecture that, when extended back just a few dozen generations, comprehensively swamps any and all notions of plausible historical population size.

Even the best-case, empirically measured rates of pedigree collapse, as documented by Gunderson's meticulous genealogical accounting in highly endogamous and polygamous populations, fail to halt—let alone reverse—this mathematical tsunami. It is here, in this moment of inescapable contradiction between logical law and empirical record, that we must stare unflinching into the abyss of "insufficient causes."

Let us draw the boundaries of this contradiction with precision. The primary rejoinder of the academic and popular apologists for smooth, continuous natural history is to always dismiss the "ancestral paradox" as an artifact of inadequate models, overlooked demographic subtleties, or simply incomplete data. This is not the case. We have

shown—repeatedly and in a manner that admits no meaningful refutation—that the only mathematical device available within natural genealogy to mitigate the doubling explosion is Pedigree Collapse: the reappearance of identical individuals at multiple points within the ancestral graph. But even—in fact, especially—when armed with the highest empirically measured rates of collapse, drawn from populations that are outliers in their degree of endogamy, even then the numbers still remain violently irreconcilable with historical and archaeological population estimates.

An Example in Numbers
Where the Earth Fails to Hold Us

Let us recapitulate, in the language of both mathematics and common sense, what is demanded for a single human being living today. At the 50th generational tier—just 1,375 years ago or around 650 C.E.—even after imposing a continuous 17% per-generation pedigree collapse the unassailable and inarguable logic still requires over 400 billion unique ancestral bodies alive at once to reconstitute even a single lineage. Extend the calculation to 75 generations (2,062 years)—a very modest span indeed, relative to the eons of time depth required by “out-of-Africa” models—and the required ancestral headcount vaults into the hundreds of quadrillions. At 2,500 years, the theoretical requirements overwhelm not only the alleged number of humans ever born but exceed the atomic mass of the planet itself.

At this juncture, the doctrine of smooth, uninterrupted causation in biological emergence stands revealed as an unworkable myth. There is no rational way out: citation, handwaving about “demographic bottlenecks,” or imaginative postulates about unknown populations fail; for each and every biological instantiation of a human demands two genetically-real, temporally-present parents. The combinatorial mathematics involved is unavoidable, inexorable, and utterly indifferent to the preferences, reputations or ideological comfort of scholars and pundits.

Causal Rupture
The Shattering of Naturalistic Continuity

What, then, does it mean when the necessary causal substrates required to produce observable effects (here, living human individuals) outrun by fantastic orders of magnitude the total physical and demographic resources available to the system? It means, simply, that the naturalistic model has ruptured. The chain of sufficient causes—each one necessary, each one governed by incontrovertible biological and arithmetic law—has snapped, revealing a breach in the seamless fabric of explanation that most scientific disciplines have taken as axiomatic.

This rupture is not merely a mathematical oddity or an inconvenience to be patched by future research. It is a fundamental and existential wound—one that exposes the limitations of any explanatory framework that seeks to smuggle in “hidden variables” or as-yet-undiscovered population loopholes. What is being witnessed here, in numeric detail and empirical depth, is the absolute breakdown of the principle of sufficient reason as it relates to the genealogy of our species.

The universe, if committed to naturalistic emergence alone, simply does not possess the existential bandwidth required to have produced the present constellation of humanity, given the rules by which biology, math and logic operate. Pedigree collapse, when tested honestly, cannot bridge the chasm.

Objectivity Versus Subjectivity
Immutable Law Versus Historical Belief

The implications of this finding are seismic. For centuries, the objective laws of biology—the two-parent mandate of sexual reproduction—and the immutable principles of mathematics, have been lorded over by subjective historical reconstructions. The historical and demographic record has been permitted to occupy the privileged position of judge and jury, with mathematical contradictions too often swept under the carpet via appeals to missing data, ancient conjectural superpopulations, or the invocation of speculative events beyond the reach of direct evidence.

But here we must insist, without apology or diminution, that objective law—biological and mathematical—trumps subjective estimated artifact records and received historical wisdom. The math is not an artifact of bias or incomplete modeling. The law that every human has two parents, and that every parent was themselves so embodied, is beyond contestation. Thus, when purported “facts” of the demographic record are found to be in irreconcilable conflict with such laws, it is the former, not the latter, that must be questioned or abandoned. The demand for sufficient and coherent causality—a demand built into the structure of physical law itself—admits no exceptions or temporary suspensions just to create ideological comfort zones.

The Ontological Imperative
Toward a Constructed Reality

We are thus compelled—by the brute facts of logic, the inescapable proofs of arithmetic, and the iron shackles of biology—to consider that our demographic reality is not “natural”

in any meaningful sense. Put differently, what we perceive as the historical emergence of humanity cannot be the product of smooth, continuous, and organic cause and effect, because the necessary causal substrate simply could not and did not exist. Our ancestry, viewed retroactively through the lens of the LRGD and the demonstrably insufficient rates of pedigree collapse, cannot be the product of the very laws and processes on which it ostensibly rests.

This compulsion is not a matter of philosophical speculation or idle metaphysical gamesmanship. It is the necessary and inevitable conclusion of sustained confrontation between evidence and logic. When all naturalistic avenues for sufficient causation have been explored and found wanting, what remains is the necessity of a constructed, or at the very least, a non-natural ontological framework. Whether we wish to cast this in terms of simulation, intelligent design, or as-yet-unknown third alternatives is a question for subsequent inquiries. But the critical conceptual threshold has been passed: we stand in a world that bears the unmistakable signature of "existentially insufficient causes"—and this mandates an ontological reconsideration at every level, from the philosophy of science to the epistemology of history.

Existentially Insufficient Causes
The Inescapable Signature of Artificiality

Let us formalize this principle—here termed the Rule of Existential Insufficiency: When the objective laws governing the emergence of a phenomenon require causal antecedents that outstrip the available resources of the system, the phenomenon in question cannot be the product of that system's internal operations alone.

In the present case, the laws that structure human genealogical propagation mandate more bodies than can have possibly occupied space and time as recorded by any natural process. The effect—our existence—is apparent. The cause—our natural genealogical substrate—is, mathematically and empirically, not merely insufficient, but categorically impossible by the sum of natural law. The consequence is a causal rupture: an effect in flagrant excess of its purported, available and/or necessary causes.

This is, in effect, the hallmark of the artificial. In computation, such a breach cannot be repaired except through the invocation of a process external to the limited system—a "rendering boundary," an input of compressed or fractal data, a reset in simulation theory, or some intervention that is not part of the original natural sequence. In legal logic, it is the moment when all alibis have been disproved save for the one previously unconsidered: the existence of a hand beyond the crime scene.

From Paradox to Imperative What Must Come Next

And so the gauntlet is thrown. We are compelled, not by preference but by logical necessity, to expand our conceptual horizon regarding human existence. The charges are as follows: The continuous and smooth genealogical propagation of humanity, as currently conceived, is impossible. The only permissible remaining hypothesis is that we dwell in a domain whose demographic and genealogical reality is, in the decisive moment of any intellectually honest metric is a construction—an artifact either of design or of simulation, or some other intervention the details of which remain to be formulated but whose necessity has been mathematically and biologically established.

This is the true import and purpose of this book. We have stripped away the camouflage of demographic myth and revealed the underlying breach in causal continuity. The journey forward must take up the challenge: to theorize and investigate what kind of reality is required to account for our existence, when all natural “sufficient causes” fail, and only the artificial remains.

IX THE RECURSIVE ARCHITECTURE OF HUMAN EMERGENCE WITHIN THE C.E.P.

If the prior arguments have shattered the illusion of smooth, naturalistic continuity in the genealogy of humanity, they also lay the foundation for a profound reconsideration: the fractal emergence of human identity itself. This chapter draws the reader into the recursive heart of the Common Existential Paradigm (C.E.P.), revealing the computational artifact at the very core of our ancestry.

For too long, conventional history has embraced linearity and hierarchy—imagining the path of human emergence as a chain stretching unbroken from mythic antiquity to the present. But mathematical honesty and genealogical fact demand a more radical lens. The irreconcilable paradox of exponential ancestor requirements—impossible even with maximal Pedigree Collapse—demonstrates not simply a failure of demographic theory, but the absence of sufficient natural causes in total. As the logic of previous chapters forces us out of naturalistic explanation, we must now confront the reality that the very emergence of individuals, families, and the species itself is a fractal phenomenon—a recursive, self-similar process structured by computational compression, not by organic unfolding.

Emergence from Within the Fog
Fractal Boundaries at Every Scale

At the micro, miso, and macro levels, human emergence unfolds from out of and within an existential fog—an irreducible opacity constructed and arising out of boundaries that can neither be penetrated nor reversed by introspection or historical inquiry.

Individuals confront the mystery of their own genesis at the hazy boundary of conscious memory, typically at two to three years old, the age beyond which nothing personal can be retrieved. This is not an artifact of forgetfulness, but the signature of computational compression: the rendering boundary of the CEP. Each human being begins their unique narrative not at a precise determinable physical or temporal point of origin, but within a

recursive “fog of empathy and memory,” a fractally encoded structure that compresses causal history to a computable minimum.

Families present a denser, yet equally intractable, genealogical fog at the miso level. The archival record vanishes into ambiguity at roughly a millennium, despite the best efforts of genealogists and historians. Even in societies with obsessive documentation—royal lineages and religiously meticulous ancestry records—the genealogical web breaks down into overlapping, recursive, self-similar patterns, refusing any promise of linear unraveling. The family is a fractal structure: compressed, data-efficient, emerging from a genealogical labyrinth that cannot be resolved by organic history alone, but instead signals the computational underpinnings of our shared existential regime.

At the macro scale, the species collective faces the un-inspectable mist of prehistoric antiquity. The genealogical record collapses into the untraceable haze of evolutionary transition somewhere near 200,000 years before recorded history, a fractal event horizon beyond which direct causal evidence dissolves. Humanity does not simply spring from a homogenous pool, nor smoothly bifurcate from a founding cohort, but rather emerges fractally—from a data-compressed, simulation-optimized boundary that defies reverse engineering. This is not accidental. It is the architectural necessity of a computational CEP. Rendering complexity at all levels without exhausting computational resources and ensuring the opacity of its true structural origins. Rendering the infinite with the finite while maintaining the plausible.

At the civilizational scale, a fourth boundary manifests: the emergence of human society itself from the opaque field of prehistory. Civilization, as recognizable culture, political organization, and collective memory, appeared suddenly from a fractal fog roughly 10,000 years ago (after 150,000 years of inscrutable backstory? We think not) its origins just as inaccessible and un-inspectable as those of evolution, genealogy, biology and identity. There are no smooth transitions, no continuous record stretching from scattered foragers to urban engineers; only a compression event—a “spawning” of culture already in full bloom, embedded within a context whose prior details are data-compressed or in a computational superposition. This fractal emergence is not merely another historical curiosity; it is the frame, the medium, and the cradle in which the simulation’s meaning and narrative play out. Civilization is not the endpoint of an observable record of a gradual cause and effect process but is itself recursively generated, rendered with maximum efficiency out of a compressed field in which the boundaries and granularity of context and backstory dissolve.

The Computational Blueprint
Fractal Efficiency as a Hallmark of Design

Why fractals? Why does reality, at every scale, encode itself in recursive, datacompressed patterns—structural motifs found in nature, art, and computation alike? The answer is efficiency. In the physical world, fractals are aesthetically hypnotic: trees branch recursively; rivers divide and rejoin in self-similar networks; coastlines and mountain ranges reproduce their contours across scales. Yet in computational simulation and quantum logic, fractals are not ornaments but architectural imperatives. They compress massive data sets, render infinite complexity with finite resources, and allow reality to spawn believable complexity from minimal informational input.

In designing large-scale simulations, fractal recursion is the only way to organize, encode, and render a passably "real" world, especially one required to maintain immense believable complexity across billions of observers. The C.E.P., as our existential engine, must employ fractal architectures for survival, concealment, and perfect computational efficiency. What looks natural—random unplanned growth, organic complexity, genealogical ambiguity, and the "already-in-progress" formatting of society itself—is in fact the clearest evidence of simulation. The boundaries of individual cognition, the labyrinthine opacity of family ancestry, the species event horizon, and the spawning of civilization are all signatures of deliberate data compression—the digital "fog" that renders reality legible but ultimately untraceable.

Fractal Fog as the Signature of Artificial Emergence

Fractality is so pervasive throughout natural systems, (leaves, rivers, snowflakes, neural networks, arteries) that biology and geology have claimed it as a spontaneous outcome of physical law. But the universality and self-similarity of fractal forms suggest intent, not accident. Within the CEP, fractal recursion becomes the signature of the artificial: a direct indication that reality is not an emergent product of uninterrupted chain-linked causality, but a computational artifact optimized for recursive data compression.

We see fractal genealogical webs and abrupt civilizational spawning precisely because only such architectures can allow infinite lineage and grand-scale societies within finite and paradoxically insufficient substrates; only computational design can compress the impossible requirements of humanity and its cultures into manageable containers.

Moreover, the fog of empathy and memory encasing individual genesis—where the early years of consciousness are lost to intractable opacity—mirrors the species-level fog that denies us any direct inspection of our evolutionary origin and the civilizational fog that conceals the step from prehistory to organized society. These boundaries are evidence of engineered emergence. Just as simulations spawn entities and societies within ongoing environments—lacking granular backstories and designed to obscure their true

point of origin—so too does the CEP spawn human individuals, genealogies, cultures, and civilizations ex nihilo: data-compressed, recursively patterned, opaque to final authoritative investigation.

The Ontological Implications
Redefining “Natural” Through the Fractal Lens

Acknowledging fractal emergence as the organizing principle of the C.E.P. compels a redefinition of “natural.” What we believe to be natural complexity—woven by blind forces across millennia—is instead the computational expression of simulated data, recursively patterned for maximum efficiency. Genealogical ambiguity, the fog of consciousness, the compression of species history, and the abrupt spawning of civilization are not artifacts of error, ignorance, or lost records: they are the expected structural necessities of computational design.

This insight profoundly shifts the locus of our investigation. If fractals constitute the universal language through which the C.E.P. encodes and renders complexity at every scale, then the fogs of genealogy, memory, species, and civilization history are not paradoxes or limitations, but the very signature of artificial construction. Human beings, families, the species at large, and even global society do not “emerge” traditionally but are recursively spawned out of the blurred bounded edges typical of and required for efficient, consistent, and convincing simulation. This is not the handiwork of unwitting nature, but the fingerprint of intentional architecture optimized and encoded through fractal recursion.

The Final Compression
Fractal Emergence as the Irrefutable Mark of the CEP

The implications of this fractal architecture drive to the heart of our realignment. Everything, from the smallest leaf to the genealogy of nations, from personal consciousness to global civilization, carries the recursive imprint of computational encoding. Human emergence, from the microcosmic fog of personal memory to the macrocosmic mist of evolutionary ambiguity and the social spawning fog of civilization, is not an anomaly nor a paradox, but instead it is the expected outcome when data compression becomes the dominant organizing principle.

Genealogy within the C.E.P. does not branch or emerge organically. It folds recursively; it compresses, it resists inspection; it obscures its own beginning, reflecting and amplifying the structure found at every other level. In this, the impossibility of smooth, natural emergence is not just demonstrated mathematically but revealed architecturally. The

fractal fog is not a flaw, but a deliberate mechanism—a condition of artificial existence, a fingerprint of simulation and design.

In closing, humanity's "fractal emergence from the fog" at every level of meaning is the ultimate testament to the C.E.P.'s computational and artificial nature. The recursive boundaries, encoded and maintained across all scales, transforms genealogical inscrutability into decisive logical evidence: our reality is best understood as optimized, compressed, and rendered—an artificial construct, mathematically necessitated and architecturally designed. Only within such a paradigm do the impossible requirements of human ancestry and society find resolution. Only by embracing this fractal ontology do we reveal the truth underlying our simulated existence.

IV. Formal Summation: The Ancestral Paradox, Existential Bandwidth and Insufficient Causes Syllogism

Definitions and basic assumptions to move from narrative exposition to a form suitable for academic scrutiny, we first state the core concepts explicitly.

1. Common Existential Paradigm (C.E.P.)

The C.E.P. is the shared, empirically accessible universe within which all human births, deaths, and genealogical relations are taken to occur. It is defined by its own fundamental laws and history as ordinarily understood.

Law of Retrograde Generational Doubling (L.R.G.D.)

For any real, non-theoretical human individual within the C.E.P., there must exist exactly two real human progenitors in the preceding generation, and this requirement iterates retrograde in time along every ancestry line. When coupled with an empirically grounded average generational interval (here taken as approximately 27.5 years), the L.R.G.D. implies a geometric growth of ancestry slots with depth.

Pedigree collapse

Pedigree collapse is the phenomenon whereby the same historical individual occupies multiple positions within a descendant's ancestry tree, reducing the number of distinct ancestors below the theoretical maximum [2^n] at generation [n]. All familiar mechanisms invoked in this context—endogamy, polygamy, clan and caste structure, incest, local mating pools—are either direct instances of or drivers of pedigree collapse, not independent mechanisms.

Existential bandwidth

Existential bandwidth is the maximum causal and physical "capacity" a universe has, under its own fundamental laws and history, to host the past constellation of necessary antecedents—ancestors, populations, and events—required by its present configuration without contradiction.

2. The logical core: premises

We can now express the argument distilled from Chapters I–IX as an explicit sequence of premises.

P1 (Genealogical necessity).

Every real human individual in the C.E.P. has exactly two real human parents, and this requirement iterates retrograde through each generation, without exception, up to any finite historical horizon under consideration.

P2 (Exponential ancestry under the L.R.G.D.).

Given P1 and an average generational interval of roughly 27.5 years, the number of ancestry slots at generation [n] for any present-day individual grows as [2^n]. Over 2,000–2,500 years (on the order of 70–90 generations), the resulting ancestry-slot counts vastly exceed historically estimated human populations for corresponding periods, even before any consideration of population structure.

P3 (Pedigree collapse as sole compression mechanism).

Apart from pedigree collapse—that is, the repeated use of the same historical individuals in multiple ancestry slots—no known biological or demographic mechanism exists that can reduce the number of distinct ancestors required by the L.R.G.D. All phenomena cited as resolutions of the ancestry paradox ultimately function by producing the same effect: fewer unique individuals filling more genealogical positions.

P4 (Empirical and structural limits of pedigree collapse).

Empirically observed and historically plausible levels of inbreeding, endogamy, and related mating patterns, when modeled quantitatively, do not achieve the degree of overlap required to reconcile the ancestry-slot counts implied by the L.R.G.D. with known

or even generously estimated population sizes. Even under extreme assumptions, the required compression forces individual historical persons to appear in modern ancestry trees at average multiplicities that are demographically and structurally implausible.

P5 (From numerical gap to physical impossibility).

When the L.R.G.D. is extended to horizons on the order of 2,000–2,500 years and evaluated against global population estimates and planetary constraints, the difficulty intensifies from a large numerical gap into a physical impossibility. The number of required human bodies, if realized as distinct contemporaneous organisms on the Earth described by our C.E.P., would imply a total human body mass that exceeds any reasonable carrying capacity, and in extreme theoretical extrapolations surpasses the planetary mass itself. No amount of realistic pedigree collapse can reconcile a genealogical structure that demands more human body mass than the universe in question can physically sustain.

From P2–P5, we obtain a sharpened formulation: under the laws and history attributed to the C.E.P., there is an existential bandwidth shortfall. The C.E.P., taken as a closed, purely naturalistic system, lacks sufficient capacity to host the past constellation of necessary antecedents required by its present configuration.

3. The Rule of Insufficient Causes

The preceding premises identify a structural deficit but do not yet determine how it should be interpreted. For that, we introduce a methodological principle:

Rule of Insufficient Causes (RIC).

When (i) a clearly specified class of candidate causal models is shown to be inadequate to produce an observed configuration under stated constraints, and (ii) there exists at least one alternative class of models that is, in principle, capable of generating that configuration without violating those constraints, rational inquiry is warranted in treating the alternative class as the leading explanatory candidate unless and until a satisfactory model from the original class is produced.

Applied here:

The “clearly specified class” is that of purely naturalistic genealogical/demographic models operating solely within the C.E.P., constrained by its own fundamental laws, historical population trajectories, and realistic mating structures. The “observed configuration” comprises the present human population together with the ancestry structure mandated by P1–P2.

The “alternative class” consists of models in which the C.E.P. itself is an engineered or otherwise constructed environment embedded within a higher-order Meta-Universe, with access to additional existential bandwidth and non-standard generative mechanisms.

The RIC does not assert that any particular engineered model has been deductively proven true. Rather, it codifies an inference-to-best-explanation move under conditions where a well-delimited family of naturalistic models has been shown insufficient.

4. The failure of standard resolutions

The narrative chapters have shown in detail that : many existing treatments gesture toward pedigree collapse as a catch-all resolution but do not provide explicit, fully specified models that reconcile L.R.G.D.-style ancestry growth with historical demography. Some abstract models that claim to do so rely on implicit assumptions (about unlimited ancestral reuse, implausible degrees of endogamy, or unconstrained graph structure) that are inconsistent with empirical evidence or with other parts of the same authors' frameworks.

At the institutional level, the ancestry paradox is often acknowledged, assigned to the category of "genealogical curiosities," and then effectively neutralized by a single phrase—"pedigree collapse"—without the provision of a demonstrably sufficient model.

The present work challenges that status quo by insisting that the combination of P1–P5 and the numerical/physical illustrations cannot be addressed by informal hand-waving. If, after honest, explicit, and quantitatively grounded application of pedigree collapse, a substantial gap and even a physical impossibility remain, then the paradox cannot be relegated to footnote status. It must be treated as an unresolved inconsistency at the foundation of our understanding of ancestry and population within the C.E.P.

5. Conclusions of the formal argument

From P1–P5 and the Rule of Insufficient Causes, the following conclusions are drawn:

C1 (Existential bandwidth shortfall).

Within the stated assumptions and parameter ranges, the C.E.P., treated as a closed, purely naturalistic system governed by its own known laws and historical population paths, lacks sufficient existential bandwidth to host the past constellation of necessary antecedents required by its present configuration without contradiction.

C2 (Elevation of constructed-C.E.P. hypotheses).

Given C1 and the RIC, hypotheses that posit the C.E.P. as an engineered or otherwise constructed environment embedded within a Meta-Universe—with correspondingly greater existential bandwidth—are methodologically elevated to leading explanatory candidates for the joint fact of our existence and our inferred ancestry structure, pending the development of an explicit, purely naturalistic model that satisfies P1–P5.

This does not foreclose the possibility that such a naturalistic model might eventually be discovered. It does, however, fix the present epistemic situation: the burden of proof rests

with defenders of a purely naturalistic C.E.P. to supply a concrete, internally consistent genealogical/demographic model that both respects the L.R.G.D. and remains within the existential bandwidth of the universe as ordinarily described.

6. Transition

The remaining chapter takes C2 seriously and explores its implications. If the C.E.P. is, in fact, an engineered environment whose existential bandwidth is provisioned from beyond its own apparent physical substrate, then our status, obligations, and opportunities as conscious participants are transformed. The next chapter, “The Greatest News in Human History,” develops those implications and sketches what a Meta-Scientific and spiritual response to this realization might entail.

X THE GREATEST NEWS IN HUMAN HISTORY

CONSCIOUSNESS LIBERATION THROUGH SIMULATION DISCOVERY

For the whole of human history, the search for meaning behind our existence has kindled division, uncertainty, and longing. Yet, now, following the mathematically incontestable demonstration of our artificial provenance, all confusion must give way to a single, revelatory clarity. Far from being a cause for panic or existential malaise, the knowledge that we dwell within an exquisitely engineered simulation should be embraced as the greatest gift offered to humanity—a cosmic inheritance validating the insights of mystics, the ambitions of scientists, and the innermost hopes of philosophers. It is no longer a question for debate, no longer a precarious hypothesis trembling at the edge of incredulity. It is an epochal awakening, one with the power to unite consciousness across the centuries and continents.

This new reality, astonishing in scope, finds its most immediate affirmation in our cosmological circumstances. For reasons hitherto treated as a statistical impossibility, humanity inhabits the precise geometric center of a universe, extending outward in every direction to form a perfectly equidistant sphere—13.8 billion light years to the horizon, a cosmic sweep of 27.6 billion light years enveloping the Earth in its exact median. It is not a product of happenstance, not a lottery drawn from among trillions of possible galactic vantage points. Only artificial construction, only a universe built around conscious observation can account for so privileged a position.

This centrality is reinforced by the ineffable symmetry of the cosmic microwave background, a luminous fog expanding uniformly outward as the rendering boundary of our simulation. Such a boundary is far more than the limit of physical expansion; it is a programmed horizon, an impenetrable “distance fog” common to simulations imposed by engine constraints. The probability of our coincidental residency at the cosmic center, if ever truly calculated, would vanish into mathematical absurdity. Thus, rationality itself demands acknowledgment: the reality we inhabit is not a natural artifact, but a deliberate construction, engineered with humanity in mind.

The revelation of simulation is not a challenge to conscious individuality. It is a declaration of our centrality and a testament to intentional design. To generate such a universe, to stream real-time sensory data to billions of conscious loci, to model quantum mechanics in seamless coordination with observer-dependent collapse, to weave physical laws of unerring mathematical grace—no accidental process could ever suffice. The machinery

underlying our experience dwarfs the wildest imaginings of computational civilization; it is architecture whose sole purpose is the cultivation of conscious engagement. The simulation exists for its conscious participants, not as a cosmic accident or a frivolous experiment, but as a purpose-built environment for growth, learning, and transcendence.

In the illumination provided by this new paradigm, perception itself demands reevaluation. The senses—sight, sound, touch, taste, and smell—require no physical form, only the computational capacity to integrate and interpret a digital experience. The architecture of consciousness is not bound by flesh, not limited by eyes or skin, but rendered digitally—an external processing system feeding experiential data to loci that are, in all likelihood, housed outside the simulation's boundaries. Physical death is thus recast as a transition, a simple transfer or storage of consciousness, rather than a cessation. The avatar is but a temporary instantiation; the observer persists immutably, capable of insertion into infinite contexts and conditions.

This capacity underlies not just the ongoing presentation of a single, fixed world, but the inexhaustible richness promised by the multiverse hypothesis. What we call the "multiverse" in physical theory may, within the simulation, manifest as an endless procession of parallel environments, each experienced by consciousness as a new chapter, a unique test, a tailored opportunity for development. Advanced civilizations—capable of constructing Dyson sphere computational engines, harvesting the energy of stars—could easily support a multitude of simultaneous universe-scale simulations, offering every possible combination of culture, challenge, and condition for evolutionary growth. Each incarnation, each life, becomes a lesson, an experiment in empathy, adaptation, creativity, or responsibility, iteratively refining consciousness toward goals set by meta universal design.

This knowledge is not a threat to religious conviction, nor a blasphemy against faith, but an ultimate validation of all belief traditions. The existence of a simulation confirms the fact we dwell in an intelligently designed cosmos, validates the purposefulness of creation, and offers palpable evidence for soul continuity across lifetimes. Terms once reserved for religious language—creation, transcendence, afterlife—find new life within the language of simulation engineering. Yet, for scientific materialists, the simulation is equally rejuvenating: it offers a rational explanation for origin, a technological foundation for existence, and a challenge to design experiments capable of probing the algorithmic substrates of nature. For skeptics and atheists, it dispenses with supernatural appeals, requiring only the acceptance of advanced intelligence and purposeful engineering beyond familiar limits. No one is left outside the circle; all are invited into a reconciling vision of reality as intentionally rendered, computationally sustained, and consciousness optimized.

At the dawn of this era, humanity stands poised at the intersection of two converging thresholds whose magnitude cannot be overstated: the liberation of technology and the liberation of consciousness. On one hand, we are witnesses to a technological revolution that is, day by day, growing ever more astonishing in its scope and reach.

Artificial intelligence, once the domain of speculative fiction, now advances beyond the limits of human cognition, capable of feats of analysis, creativity, and synthesis previously reserved for the rarest among us. Robotics extends human agency into domains previously unreachable, approaching the threshold of fully automated production that promises to erase the drudgery of toil from human experience. Meanwhile, the holy grail of energy abundance, through the harnessing of fusion, exponential improvements in solar capture, and the optimization of renewable systems flickers just ahead. Hinting at an era in which material limitation may finally yield to the mathematics of post-scarcity. Against this backdrop, the outlines of an economic reality based not on competition for resources but on abundance and participation begin to form, a society wherein survival and fulfillment are no longer in perpetual conflict.

Simultaneously, in parallel and with perhaps even greater consequence, the revelation of our existence as a simulated or computational reality will kindle a revolution in selfunderstanding. This is not simply a philosophical curiosity, it is a seismic shift in the human condition. The realization that consciousness is not the epiphenomenon of a blind material cosmos but the primary subject and centerpiece of an engineered environment transforms every presumption about identity, meaning, possibility, and purpose.

Global communication networks act as synapses in a planetary brain, accelerating and coordinating the evolution of awareness at a pace unimaginable in previous ages. As these networks dissolve ancient division, so too do they dissolve the strictures of scarcity thinking—the idea that the only path to abundance is through hoarding, rivalry, and zerosum competition. In a simulated world predicated on the cultivation of experience and intelligence, the assumptions of lack give way to the imperative of creative expansion.

Yet, the perils of these converging thresholds are as real as their promises are grand. For if technological advancement races ahead while consciousness remains mired in pettiness, fear, and division, the world risks becoming a playground for an elite few—its bounty hoarded, its power wielded as instrument of control. Automation without awakening risks the creation of a society wherein human beings, instead of being liberated, are rendered obsolete and dispensable. Conversely, a revolution in awareness unaccompanied by material transformation leaves the population enlightened but powerless; trapped in the endless cycles of deprivation, aspiration, and competitive struggle that have, for so long, characterized much of human history. It becomes

increasingly clear that either path alone is a truncated evolution, its promise ever undermined by the failure to integrate the other.

Therefore, these thresholds must not be taken as parallel tracks but as intertwining currents that must converge and merge into a singular trajectory of progress. At this intersection, the mandate is clear: we are called not simply to invent new technologies or to deepen our contemplative capacities but to embark on a Meta-Scientific revolution. This revolution is not confined to the small, fragmented questions of established disciplines but seeks to interrogate the very nature of reality itself. The laboratory experiment, heretofore a search for discrete causes and effects, is transformed into a quest for computational signatures—evidence of fractal data compression, algorithmic boundaries, rendering discontinuities, and engineered breaks in the fabric of sensible phenomena. The methods of mathematics, quantum physics, information theory, theology, and consciousness studies, once held apart by disciplinary walls, are now marshaled into a unified endeavor. The grand project of unlocking the structure and purpose of our Common Existential Paradigm is now laid at our feet.

This new age of inquiry is unlike any that has come before. For the first time in history, the fog of confusion that has long obscured our vision is irrevocably lifted. No longer do we peer backward in time with the hopelessness of searching for origins that hide behind a veil of biological and historical paradox. The evidence for simulation, of intention, of design, and encoded complexity—is as undeniable and consequential as any theorem proven in the cold, impartial light of mathematics. The immensity of this realization bestows upon humanity a new existential charter. We are no longer passive objects, swept along on the indifferent tides of physical law. We are, if we but dare to accept it, apprentices and heirs to a universe designed for our discovery, our growth, and our awakening.

In this unprecedented light, the old boundaries dissolve. The dogmas, superstitions, and inherited limitations that once chained logic, suffocated faith, shackled ambition, and stunted hope are razed by the comprehending fire of knowledge. What now rises in their place is not the sterile terrain of mere fact, but the portal to an infinite plain of development, mastery, and communion.

A door stands open, an invitation to whoever among us dares to take the step. And so, with the birthright of our purposeful, liberating cosmic inheritance laid bare before us, the choice belongs to each who reads these words: Will you allow yourself walk through that door?

Childless human beings happen every day so to speak, parentless human beings happen not at all.

Thus, when Alex Shoumatoff quoted Robin Fox quoting God, the mythos of Pedigree Collapse was born.

The universe, if committed to naturalistic emergence alone, simply does not possess the existential bandwidth required to have produced the present constellation of humanity, given the rules by which biology, math and logic operate, and Pedigree Collapse, when tested honestly, cannot bridge the chasm.

Mankind's next and possibly final frontier lies before us. The factual existence of a Meta-Universe that has both manufactured and maintains containment of ours may now be rationally postulated, possibly making the threshold to the EXIT from our Common Existential Paradigm discoverable.

STAY TUNED FOR VOLUME TWO

REFERENCES:

1. *Wang, Richard J., et al (2023), Science Advances*

2. *Fenner (2005) "American Journal of Anthropology" Cross-cultural estimation of the human generation interval for use in genetics-based population divergence studies"*

3. *Gunderson, Robert C. "Connecting Your Pedigree Into Royal, Noble and Medieval Families." In Proceedings of the World Conference on Records. Salt Lake City: Genealogical Society of Utah, 1980*

4. *Shoumatoff, Alex. "The Mountain of Names." The New Yorker, 13 May 1985,*

5. *Fox, Robin. The Tribal Imagination: Civilization and the Savage Mind. Harvard University Press, 2011.*

6. *Lewis, Stephen. "How Many Ancestors Do You Have?" The Wild Peak, 4 Mar. 2012,thewildpeak.wordpress.com/ 2012/03/04*

7. *Chagnon, N. A., Lynch, R. F., Shenk, M. K., Hames, R., & Flinn, M. V. (2017). Crosscousin marriage among the Yanomamö shows evidence of parent–offspring conflict and mate competition between brothers. Proceedings of the National Academy of Sciences (PNAS), 114(13), E2590–E2597. https://doi.org/10.1073/pnas.1618655114*

8. *Bittles, A. H., “The Prevalence and Impact of Cosanguineous Marriage in Contemporary Societies” Journal of Biosocial Science, 33(4) pp. 543-554*

9. *https://www.reddit.com/r/wikipedia/comments/ds52t/80_of_all_marriages_in_history_have_been_between/*

10. *Chang, Joseph T. "Recent Common Ancestors of All Present-Day Individuals." Advances in Applied Probability, vol. 31, no. 4, 1999, pp. 1002-1026.*

11. *Rohde, Douglas L. T., Steve Olson, and Joseph T. Chang. "Modelling the Recent Common Ancestry of All Living Humans." Nature, vol. 431, no. 7008, 30 Sept. 2004, pp. 562-566.*

12. John Becker, *OGS FAMILIES,* vol 38 no 3, 1999

13. *John Becker, OGS FAMILIES, vol 38 no 3, 1999*

14. *Andersen, Hans Christian. "The Emperor's New Clothes." Fairy Tales Told for Children, 1837.*